When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

A speech model's decision to speak does not establish that it can answer correctly. We study whether its frozen hidden states expose a useful signal for selective escalation to a stronger expert. On MiniCPM-o 4.5, a final-layer, last-token failure probe inverts on held-out math queries with text input (AUC 0.372), whereas the same read at layer 22 reaches 0.931. A text-calibrated mid-layer probe also transfers to speech-input states (AUC 0.855). We use this read location in a separately calibrated answer-onset gate and evaluate its decisions in the model's native listen/speak protocol. On 240 internal test queries, answer-content accuracy increases from 40.8% to 62.9% at 51.7% escalation, exceeding the reported random-mixture reference by 6.8 points. Across five external speech-QA pools, the aggressive tier improves macro accuracy from 59.4% to 80.9%, versus 75.2% for the reference. Reconstructed timing diagnostics expose substantial expert-path costs; these experiments do not establish an end-to-end conversational latency benefit. The results support mid-layer competence readout as a routing signal and identify the timing, calibration, and relay constraints that must be resolved for real-time use.

1 Introduction

Spoken interaction places a tight deadline on response selection [Stivers et al., 2009]. A local speech model can maintain the conversation, while a stronger expert handles difficult questions. Calling the expert on every turn increases cost and waiting time; selective escalation instead requires an early estimate of whether the local answer will fail. A duplex model already decides when to speak, but this floor-management decision need not reveal whether it knows the answer.

Hidden-state probes and self-evaluation can estimate correctness in text models [Kadavath et al., 2022, Azaria and Mitchell, 2023, Kossen et al., 2024], and text routers allocate queries between models [Ong et al., 2024, Varshney et al., 2026]. Applying these ideas to speech introduces three questions. Does a readout distinguish failures on unseen query pools? Does it survive the move from text to audio input? And where in the streaming protocol is the state actually available? High in-distribution AUC or a fast offline probe alone does not answer these questions.

We investigate them on frozen MiniCPM-o 4.5 [Cui et al., 2026], with raw-backbone and speech-model controls. The representational result is input- and position-specific: on text input, a final-layer, last-token probe falls below chance on held-out math, while a mid-layer probe remains predictive. The final-layer failure does not generalize to audio-calibrated, audio-input probes. The practical motivation for the middle layer is therefore robustness across the tested conditions, rather than a universal absence of competence information at the output.

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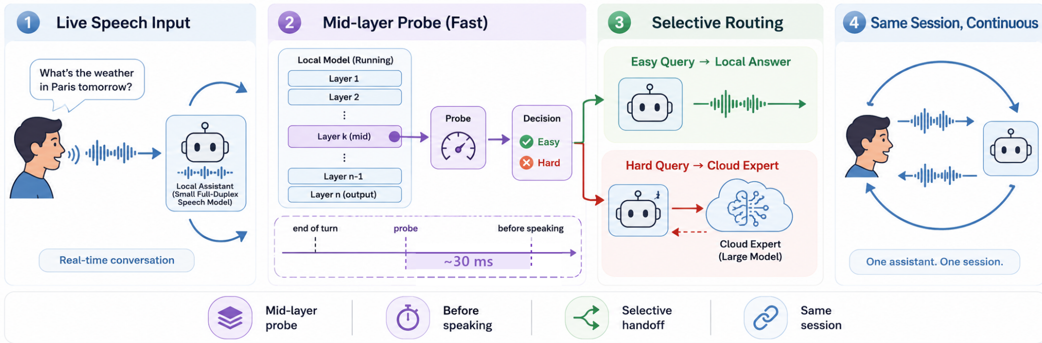


Figure 1: **The deployed system.** **(1)** Speech streams in 1 s chunks into a full-duplex local model that decides for itself when to speak. **(2)** At that speak commit, a mid-network (L22) linear read scores the turn in $\sim$30 ms, before the answer is produced. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert while the talker stalls. **(4)** The returned answer is spoken verbatim in the talker's own voice. Delivered accuracy on the internal test set rises from 40.8% to 62.9% at 52% escalation in native full-duplex sessions.

Our contributions are threefold. First, a layer and position audit shows that LOPO math AUC increases from 0.372 at the final layer to 0.931 at layer 22; matched controls constrain, but do not identify, the cause of the late-layer degradation. Second, a text-calibrated mid-layer probe transfers to speech-input states at AUC 0.855, separating representation transfer from the additional calibration needed for native serving. Third, a native-protocol benchmark shows that a separately fitted answer-onset gate selects useful expert calls: it improves internal answer-content accuracy by 22.1 points and exceeds the reported random-mixture reference by 6.8 points at the aggressive tier. We report accuracy together with the recorded timing costs and distinguish this benchmark from a causal streaming evaluation of audible responses.

2 Related Work

Competence readout. Correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024], and uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]. Intermediate-layer representations can expose information missing from model outputs [Orgad et al., 2025, Mahaut et al., 2024]; Semantic Entropy Probes amortize uncertainty estimation into a hidden-state read [Kossen et al., 2024]. Our question is how readout location, input modality, and serving protocol affect this signal after speech and duplex fine-tuning.

Selective routing. FrugalGPT, HybridLLM, and RouteLLM allocate work across language models [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024]; AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Pre-generation activation routers and frozen-state success probes are especially close to our method [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear failure probe is thus not itself the novelty. We test its robustness in speech-model representations and use a threshold to expose a tunable expert-call budget. Predicting local failure remains distinct from predicting the benefit of a particular expert.

Native duplex control. Native spoken-dialogue models coordinate listening and speaking [Défossez et al., 2024, Cui et al., 2026], using state or synchronization mechanisms [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024] or frozen-backbone adapters [Wang et al., 2024b]. BayLing-Duplex [Fang et al., 2026] makes dialogue-state decisions within a single autoregressive LLM by interleaving user audio, assistant text, and assistant audio and introducing four state tokens. Starting from GLM-4-Voice, it learns turn-taking and interruption through supervised full-duplex fine-tuning and timing-focused preference optimization. Its evaluation measures spoken-answer quality and interaction timing, providing a concrete example of learned native floor control.

Figure 2: **Text-input transfer depends on layer and position.** LOPO math AUC across layers: late last-token reads deteriorate on the duplex checkpoints, while mid-layer reads and raw-backbone controls remain predictive. Layer 22 is the MiniCPM-o 4.5 read location used subsequently. This plot does not show an equivalent collapse under audio calibration and audio input.

Competence-aware handoff. DuplexOmni, chronological thinking, and MoshiRAG connect speech to reasoning or retrieval [Huang et al., 2026, Wu et al., 2026, Chien et al., 2026]. Our work studies a decision complementary to native floor control: given a duplex-trained checkpoint, can its hidden states predict local-answer failure and guide escalation to an expert under a tunable call budget? We fit a small readout without updating the speech model. This requires regime-specific calibration and does not assume that a learned turn-taking signal predicts answer correctness.

3 Reading Competence from Hidden States

We first study failure prediction independently of the expert path. The representation experiments use 600 queries in five fixed pools, with a 360-query calibration split and a 240-query test split (§4.1). Logistic probes read stored activations from frozen checkpoints. Higher scores predict local-answer failure.

3.1 Distribution transfer exposes a readout failure

The final-layer probe appears useful in distribution (AUC 0.822), but pool identity alone reaches 0.715, and the hidden state predicts source pool at 95.8% accuracy (Appendix C). We therefore emphasize leave-one-pool-out (LOPO) transfer rather than in-distribution performance alone. With text input, the final-layer, last-token read falls to AUC 0.372 on held-out math. Query-surface routers also depend strongly on pool structure (Appendix F).

Verbalized self-evaluation has different failure modes. Pre-answer $p(\mathrm{True})$ detects the held-out trap pool (AUC 0.945), where the final-layer probe fails. Post-draft $p(\mathrm{True})$ reaches 0.899 in distribution but requires a completed draft, and can endorse confidently wrong answers. Table 5 reports these signals under their respective read conditions.

3.2 Layer and position matter

A layer-by-position sweep localizes the text-input failure to late last-token states (Figure 2). Mean pooling degrades less, and a last-token probe at layer 22 reaches **0.931** on the same held-out math pool. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96; the Qwen2.5 control family shows a similar ordering (Table 6). These comparisons associate the degradation with the tested fine-tuned checkpoints. They do not isolate a causal training objective.

The input condition is essential: with audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We retain L22 for its robustness to shifted text queries and modality transfer. Mechanism tests find neither a shift in the degradation layer under speak mode nor alignment of the inverting direction with explicit listen/speak directions (Appendix E). The layer sweep establishes a useful read location, while the mechanism remains open.

3.3 Text-to-speech representation transfer

A probe trained on text-input hidden states and evaluated on speech-input states reaches AUC **0.855** at L22. Transfer stabilizes around the middle of the network, replicates on MiniCPM-o 2.6 and Qwen2.5-Omni, and is substantially weaker in the frozen-backbone Freeze-Omni adapter control. The mid-layer pattern also appears on 200 SD-QA human recordings. Conversely, audio-input verbal self-evaluation degrades on the two MiniCPM duplex generations; adding the question transcript restores it (Appendix C). These are representation-transfer results. The native escalation gate below uses a larger, audio-derived calibration set and three positions at L22.

3.4 From a failure score to an answer-onset decision

For native serving, we concatenate the last tail state, the mean of the tail states, and the mean input state at L22 into $\phi(x) \in \mathbb{R}^{12288}$. A logistic probe gives

$$s(x) = \sigma\big(w^\top \phi(x) + b\big)\,, \qquad d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\}\,\mathbf{1}\{a(x) \geq \tau_a\}, \tag{1}$$

where $a(x)$ is a separately fitted dialogue-act score that filters floor-management turns, and τ_ℓ is a language-specific failure threshold. Calibration-score quantiles define three nominal budgets (15/30/50%); realized rates can differ on a new pool. The model weights stay frozen, but both heads require fitting.

The native capture occurs when a generation call first returns a listen-to-speak transition. It can include the first generated text unit, so we call it *answer onset*. This differs from the earlier turn-based prefill read, whose measured P50 availability was 20 ms for text and 45 ms for audio on an H100. A separate turn-end replay reported about 30 ms (Appendices G and H); those measurements are not native-path end-to-end latency. Later native reads can improve reasoning-failure prediction, at the cost of waiting for answer chunks (Appendix P.1).

4 Experimental Setup

4.1 Models, data, and calibration

The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Matched controls include raw backbones, a second MiniCPM generation, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni (Table 4). Representation analyses use bf16, greedy decoding, and seed 42. The native benchmark uses sampled decoding under the serving configuration ($T = 0.7$, top-$k = 20$); these are different experimental regimes. The escalation expert is `gpt-5.5` with low reasoning effort.

The internal 600-query set contains SimpleQA traps [Wei et al., 2024], MMLU-Pro knowledge questions [Wang et al., 2024c], TriviaQA facts [Joshi et al., 2017], Dolly/Alpaca chat prompts [Conover et al., 2023, Taori et al., 2023], and GSM8K/MATH problems [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]. Its original 360/240 split and test composition remain fixed. LLM judges score answer adequacy. A stronger re-judge of the representation-study labels agrees on 96.2% of text and 94.5% of audio cases; this agreement is not a validation of speech delivery.

Three calibration regimes must be distinguished: 360 examples for single-position representation probes; 2,310 for the earlier turn-based three-position gate; and 5,228 for the native gate, refitted on native features and judged native-answer outcomes. The native fit contains 4,986 core and 242 fresh training rows; thresholds use five-fold out-of-fold scores from the core rows, grouped by language. Neither fresh-row scores nor external evaluation-pool quantiles enter these thresholds. Seven fitting rows overlap external questions. A 5,221-row deduplicated artifact is available, but was not used in the recorded benchmark; its reported offline effects are small (Appendix A).

4.2 Native benchmark protocol and metrics

The MiniCPM upper block of Table 1 evaluates the internal test split and five external speech-QA pools: Speech TriviaQA, Speech Web Questions, Speech Llama Questions, and Mandarin Speech Reasoning QA from OpenAudioBench [Li et al., 2025], plus SD-QA recordings from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. VoiceBench AlpacaEval is reported in the table caption on its separate 1–5 scale. External audio comes from the benchmark releases; internal questions use a fixed

Table 1: Main results: selective escalation on the internal test set and five public speech-QA benchmarks (delivered answer accuracy, %, native full-duplex sessions, one live arm per tier). The probe is fixed; tier thresholds are label-free calibration quantiles per language (the internal tiers realize 8/25/52% escalation; realized rates per pool are in Appendix K). Matched-rate random mixes the always-local and measured always-escalate arms at each pool's realized rate. avg and Δ (average gain over always-local) cover five external pools in the live block and four in the NVDA block. The relay-free bound scores the expert's returned text directly. The bottom block evaluates our latest three-layer NemotronLabs-VoiceChat-11B probe by re-mixing local and cached expert outcomes after replay through its native frame protocol (Appendix D). External columns retain cached local outcomes and the prior 2,481-row fit. The Internal column judges the same pass-3 answers on all 240 queries and uses a 2,258-row refit that excludes these test IDs (with cached pass-2 training labels). All policies leave the 17 queries without an answer-onset read local, so the 15/30/50% scoreable-query budgets realize 14/28/47% overall. This Internal result is post-selection because probe architecture selection included this split. AlpacaEval, reported separately on its 1–5 scale, changes from 3.68 to 4.01 at 44% escalation on MiniCPM (matched random 3.91; always-escalate 4.20 against 4.95 for the expert text, a spoken-length cap); the pass-3 NVDA ranking changes from 3.55 to 4.51 versus 4.23 for random.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	Reason. zh	avg	Δ
native full-duplex live sessions (delivered accuracy)								
always-local	40.8	62.8	52.8	82.4	48.0	51.0	59.4	—
+ gate, conservative	46.7	60.4	59.2	78.8	54.0	62.4	63.0	+3.6
+ gate, balanced	54.2	70.0	62.4	82.4	65.0	66.3	69.2	+9.8
+ gate, aggressive	**62.9**	**88.4**	**75.2**	**85.6**	**82.5**	**72.8**	**80.9**	**+21.5**
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	67.5	75.2	+15.8
always-escalate (measured)	70.4	96.0	79.6	91.6	87.0	83.7	87.6	+28.2
expert text as returned (relay-free bound)	76.5	96.4	82.3	93.2	88.5	86.1	89.3	+29.9
NVDA three-layer answer-onset probe (pass-3 native-frame replay)								
always-local	19.6	35.8	37.6	70.8	31.0	—	43.8	—
+ gate @15%	31.7	48.4	48.8	80.7	42.0	—	55.0	+11.2
+ gate @30%	44.6	61.0	59.2	85.8	54.0	—	65.0	+21.2
+ gate @50%	60.4	76.8	69.6	89.7	66.5	—	75.7	+31.9
matched random @50%	53.5	65.7	60.2	82.0	60.0	—	67.0	+23.2
always-expert (ceiling)	91.7	95.5	82.8	93.1	89.0	—	90.1	+46.3

TTS rendering. Each query has a separate native-protocol session for each arm: local-only, three gate tiers, and always-escalate. There is one recorded run per query and arm, without repeated-run averaging.

Audio enters the local model in one-second chunks. At an escalation, transcription and the expert produce a text answer, which is cleaned and capped at 400 characters for teacher-forced TTS relay. We report *answer-content accuracy*: the judge sees local generated text or the intended relay text, not a transcription of played audio. The benchmark disables local audio generation. Uplink transcription uses the final 30 seconds of the full prerecorded question: it can access unconsumed audio at early onset and loses the beginning of longer questions. Thus the experiment evaluates routing under this input protocol, without establishing strictly causal interaction.

For the MiniCPM block, the reported random reference is

$$A_{\text{mix}}(r) = (1-r)A_{\text{local}} + rA_{\text{always}},$$

where r is the gate's realized escalation rate. This follows Table 1's convention of treating the always policy as the full-escalation endpoint; it is an analytical mixture, not a separately run randomized arm. Non-commits and a sensitivity check using the actual always-arm rate are reported in Appendix A. Timing is reconstructed from chunk offsets and logged stage durations as time to first audio: the relay path ends when its synthesized waveform is ready (expert wait plus blocking TTS), excluding the waveform's playback duration; the local path ends at text generation. Figure 3 reports mean, P50, and P95 of this diagnostic, not a client-audible completion measurement (Appendix A).

5 Escalation Results

5.1 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports answer-content accuracy on the fixed 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive.

Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix K). Aggressive exceeds the table's random reference of 56.1% by 6.8 percentage points. Per-tier internal comparisons are in Table 13. Aggressive recovers 74.6% of the measured local-to-always accuracy gap while using 51.7% expert calls. This measures selection value relative to the recorded endpoints.

Gains vary across the fixed internal strata: factual QA improves by 52.5 points, while math accuracy is unchanged. Of 60 knowledge questions, 43 exceed the expert's input window. These descriptive results identify an input constraint and task differences, without attributing the accuracy gap to truncation or changing the test mixture (Appendix A.1).

The timing diagnostic makes the cost visible (Figure 3). Internal mean time to first audio increases from 5.7 s locally to 8.7 s at balanced and 14.4 s at aggressive; aggressive P95 is 60.8 s. The always arm averages 17.6 s (P50 11.5 s). The expert call itself accounts for 6.0 s of that mean; the rest is the blocking, non-streaming synthesis of the expert text, whose tail is driven by code and markup voiced verbatim (Appendix A). A lower call rate can reduce expert-path exposure, but these tails do not support calling the current aggressive setting low latency. An earlier cached-outcome sweep suggested nominal 15/25/40% as candidates; those settings require new runs and are not substituted for the measured 15/30/50% recipe here (Appendix A).

Table 1's measured always arm and expert-text row score different objects. On the 238 internal always-arm rows with expert scores, returned text reaches 76.5% and intended relay content reaches 71.0%; the full 240-row always arm scores 70.4%. This paired 5.5-point gap combines relay content changes with judge variability; it does not isolate formatting loss or acoustic intelligibility. The table's "relay-free bound" is a returned-text reference, not a proven upper bound on every routed or spoken answer.

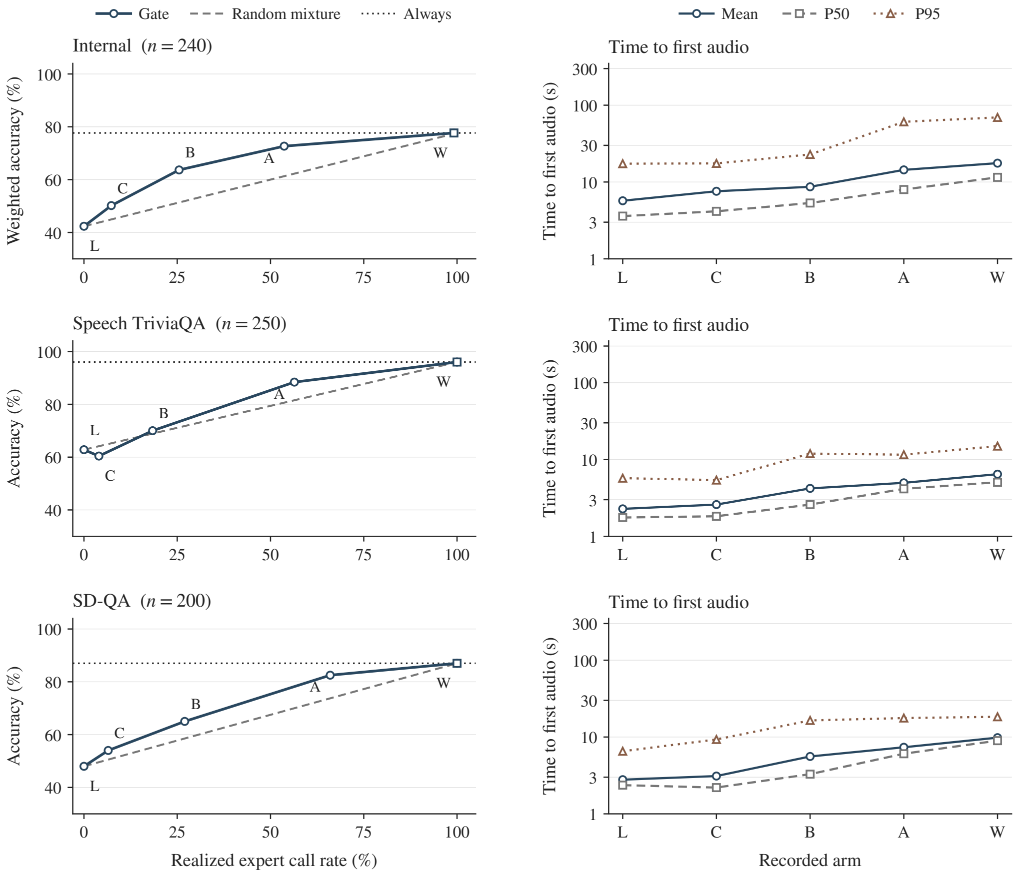
5.2 External transfer and its limits

The native gate's weights, feature definition, layer, and per-language thresholds stay fixed across the five external QA pools. The upper block's *avg* excludes the internal set; Δ is its gain over always-local. Aggressive macro accuracy rises from **59.4% to 80.9%**, compared with 75.2% for the reported random mixture: +21.5 points over local answering and +5.7 over random. Its realized call rate ranges from 18.8% on Llama Questions to 66.0% on SD-QA, so this is not a uniform 50%-budget comparison. TriviaQA and SD-QA illustrate large improvements over local answering (+25.6 and +34.5 points) and over the corresponding random reference (+6.9 and +8.8 points).

The lower tiers are less reliable. Conservative accuracy falls by 2.4 points on TriviaQA and 3.6 on Llama Questions; balanced leaves Llama Questions unchanged. Separately sampled sessions and limited expert headroom constrain the interpretation of small differences. The results support aggressive-tier gains across these pools, not improvement on every benchmark at every budget. On AlpacaEval, the table caption reports 3.68 locally and 4.01 at aggressive, with a random-mixture reference of approximately 3.9; the always arm scores 4.20 versus 4.95 for returned expert text. The relay cap is particularly restrictive for long, open-ended answers.

Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC increases from 0.755 at commitment to 0.791 after two answer chunks. This is consistent with some useful information emerging during generation, without proving a causal mechanism (Appendix P.1).

The lower block of Table 1 reports the exploratory NemotronLabs-VoiceChat-11B (NVDA) study. Its *avg* covers four English QA pools, with no internal or Mandarin entry. At a 50% replay budget, average accuracy rises from 43.8% to 75.7% ($\Delta = 31.9$ points), compared with 67.0% for random. This is native-frame replay with cached outcome mixtures, not another set of MiniCPM native sessions. The L26/L30/L34 heads reach 0.818 out-of-fold AUC and 0.816 mean external AUC in the separate probe analysis (Appendix D); Table 1 reports accuracy rather than AUC. External pools were inspected during variant review, so independent confirmation is needed.



Internal accuracy: post-hoc weights of 0.25 for knowledge and math, and 1 for other categories.
Internal timing uses the original mixture.

Figure 3: **Accuracy gains and timing costs on three pools.** Left: observed mean answer-content accuracy against realized expert call rate; C/B/A denote conservative/balanced/aggressive. The dashed line uses the original Table 1 random-mixture convention (§4.2). The horizontal always-arm reference passes through the measured always point; it is not an oracle bound. Right: mean, P50, and P95 of the reconstructed time to first audio on a log scale. L/W denote local/always. Each point uses the original recorded pass, with one run per query and arm; no repeated-run uncertainty is implied. Local rows end at answer-text completion and escalated rows when the relay waveform is synthesized; playback duration is excluded (§4.2). All five external QA pools, including the weaker cases, appear in the upper block of Table 1.

6 Discussion and Limitations

The strongest finding is representational: a mid-layer read remains predictive under conditions where the conventional final-layer, last-token probe fails. This behavior is specific to the tested checkpoints and input conditions. The fitted readout can guide escalation without updating the speech model, but the native result requires in-regime audio features and labels; it is not a zero-shot deployment of the text-calibrated probe.

Failure prediction is also only one part of useful handoff. The expert must repair the selected failures, and the serving path must preserve its answer within an acceptable wait. The current results show answer-content gains alongside long time-to-first-audio tails. The native capture can already contain generated text. In the main recorded protocol, the expert's 30-second input window admits unconsumed audio at early onset and truncates longer questions; text-based judging does not establish what listeners receive. The subsequent v2 validation preserves the consumed prefix and scores transcribed output audio (Appendix A.2); a full streaming evaluation still needs substantive speech onset and completion measured on a common clock. Floor-management and interruption checks in the appendix remain small scripted studies, without human ratings or broad multi-turn validation.

The internal mixture and its test split remain fixed. We report all five external QA pools rather than changing test composition after observing outcomes. Known calibration overlaps are disclosed; the small offline effect of removing them does not make the recorded benchmark overlap-free. Base-model pretraining contamination is unresolved. Layer selection is not nested within every LOPO fold, LLM judges have residual error, and the main table's single pass does not quantify run-to-run reliability; later internal repeats provide a limited diagnostic (Appendix A.2). The next evaluation should lock a deduplicated calibration manifest and validation-selected thresholds before evaluation on fresh, audited benchmarks.

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A Current benchmark protocol and evidence scope

Table 1 retains the original main-results presentation: its upper block reports MiniCPM native-benchmark accuracy, while its lower block reports NVDA native-frame replay. Figure 3 is recomputed from the committed MiniCPM judged rows with the TTS relay and follows the upper block's random-mixture convention. The 35 input parquets match the hashes in the accompanying benchmark audit. Steer-relay runs and later patched-model variants use different protocols and are not pooled with these arms. The remaining appendix preserves representation studies, retired turn-based experiments, concurrent-serving controls, and exploratory native variants. Their settings and denominators differ. They are not additional repeats of the main benchmark, and historical "delivered" or "latency" labels must be interpreted using each experiment's scored object and timer endpoint.

Calibration ledger. Representation probes use 360 calibration queries from the original 600-query mixture and single-position 4,096-dimensional features. The earlier turn-based gate uses 2,310 calibration rows and three positions. The main native gate uses 5,228 rows, native-answer labels, three L22 positions, and logistic regression with $C = 3 \times 10^{-4}$. The native merge has 4,986 core rows and 242 fresh training rows. Five-fold stratified cross-validation uses shuffle seed 42. Thresholds are the $(1 - r)$ quantiles of core-row OOF scores within each language group at nominal budgets $r \in \{0.15, 0.30, 0.50\}$. Fresh training rows participate in the fit but not the quantile base. The released OOF scores yield AUC 0.8477 on 5,228 rows and 0.8493 on the 5,221-row refit after excluding seven overlapping IDs. Recomputing the 12 language/tier/fit quantiles from these scores reproduces both artifacts' thresholds to four decimals. This checks the exported scores and threshold calculation, not an independent feature-level refit. The recorded arms use the original artifact and 15/30/50% tiers; neither the deduplicated gate nor the proposed 15/25/40% tiers was used for these results. The 240 internal test queries retain their original membership and pool proportions.

Capture and input availability. The native feature capture follows a generation call returning the first listen-to-speak transition. Thus the tail can include the first generated text unit. Earlier prefill timing measurements concern a different read point. In the benchmark implementation, the expert receives the last 30 seconds of the preloaded question waveform, rather than a prefix clipped to the current input chunk. This admits future audio when the model commits early. On the internal set, escalated early commits occur in 1/240, 10/240, and 18/240 sessions for conservative, balanced, and aggressive, and 26/240 for always (26 of the 238 fired sessions). Always-arm early escalations also occur in 38/199 AlpacaEval and 18/202 Mandarin reasoning sessions. Conversely, 46/240 internal question waveforms exceed 30 seconds. The always arm fires on 45 of these, dropping their beginnings from the expert input; the remaining query does not trigger a request. The input rule follows from the archived code; the exact request audio was not persisted. These data therefore cannot establish a causal streaming latency benefit. Preserving the received question prefix may change accuracy and waiting time and requires new runs.

Scored object and denominators. The judge receives generated local text or cleaned intended relay text. Local generation uses `generate_audio=False`; relay synthesis records a duration, but no local/relay audio or client playback timestamps are archived. The field named `transcript` is uplink ASR of the user's question, not transcription of an answer. The upper block's "delivered accuracy" therefore means answer-content accuracy. Each arm has 240 internal rows. The always policy commits and escalates on 238 (99.2%); its two non-commit rows remain in the denominator. Expert-text scores exist for 238 rows, and returned-text/relay comparisons use this matched subset. The WebQ always arm similarly realizes 99.2% calls. The original Table 1 reference mixes local and always accuracy with weight r, the gate's call rate, treating the always policy as the 100% endpoint. It gives internal aggressive random accuracy 56.1% and a 6.8-point gate advantage. Normalizing instead by the observed always-arm rate ($\lambda = r/r_{\text{always}}$) gives 56.2% and 6.7 points. We report this as a sensitivity check, while the main text and Figure 3 use the original table's convention. For AlpacaEval, mixing the rounded caption anchors (3.68, 4.20) at the rounded 44% rate gives 3.91; using all unrounded row values gives 3.9047. Both support the main-text description of approximately 3.9, on a scale separate from binary QA accuracy.

Timing reconstruction. Let $o_i = \max(0, c_i + 1 - n_i)$ be the chunk offset relative to question end, where c_i is the zero-based onset chunk and n_i the number of chunks. Missing onset uses $c_i = n_i$, reproducing the historical plotting convention. The recorded diagnostic is the time to first audio

(TTFA),

$$\widetilde{T}_i = \begin{cases} o_i + t_i^{\mathrm{answer}}, & \text{local}, \\ o_i + t_i^{\mathrm{stall}} + w_i + t_i^{\mathrm{synth}}, & \text{escalated}, \end{cases}$$

where durations are in seconds and w_i is the count of waiting chunks (the expert call runs during these chunks; w_i tracks the logged expert latency with correlation 0.96–0.99). t_i^{synth} is the wall time of the blocking, non-streaming teacher-forced TTS call that voices the expert text; the relay waveform can start playing when it returns. The waveform's own duration d_i^{relay} (mean 27.8 s on the always arm, 61% of the sum when included) is playback, not latency, and is reported as the "play" column of Table 14; an earlier draft of this table added it to the escalated rows, which is why its always arm read 45.1 s. We retain the remaining definition to expose the recorded cost profile, not as a synchronized wall-clock identity: asynchronous stages can overlap, missing fields are zero-filled by the plotting recipe, and the two paths end at different output events. Local t_i^{answer} ends at text generation, an upper bound on the duplex talker's first audio; client-audible completion is not measured. Figure 3 reports the mean and within-arm P50/P95 over queries in the original recorded pass. These quantiles do not measure run-to-run uncertainty, and query bootstraps in historical figures are not substitutes for repeated executions.

Operating-point candidates, not replacement results. The archived sweep summary records a cached local/always-outcome analysis (4 September 2026), shown in Table 2. Nominal 15/25/40% is a candidate recipe for confirmation. Within that same remix, reducing the nominal aggressive budget from 50% to 40% changes mean reconstructed time (under the sweep's playback-inclusive definition) from 23.3 to 19.6 s (15.9% lower), with accuracy changing from 64.2% to 60.4%. Comparing 19.6 s from this remix directly against the separately recorded aggressive arm (14.4 s TTFA; 27.3 s under the same playback-inclusive definition) would mix protocols. The historical sweep's threshold grid differs from both released core-OOF grids: at nominal 30% in Mandarin, it uses 0.7516, versus 0.3862 for the deployed artifact and 0.3695 for the deduplicated artifact. The revised export identifies this grid as historical, with its unavailable fit metadata explicitly left unspecified; it separately supplies a fresh grid from the deduplicated core OOF scores. The historical sweep uses already inspected test outcomes; it provides exploratory operating points, not independent validation or a deployed threshold change.

Table 2: Reported internal cached-outcome sweep. Nominal budget refers to calibration quantiles; accuracy and mean time are remix estimates. New native runs are required before replacing the main operating points.

Nominal budget (%)	Realized rate (%)	Accuracy (%)	Mean diagnostic (s)
15	8.8	44.6	9.5
20	14.2	47.9	13.1
25	21.7	52.5	14.4
30	27.5	55.4	16.0
40	40.4	60.4	19.6
50	50.0	64.2	23.3

Overlap audit and unresolved contamination. The recorded normalized-exact/token-Jaccard (≥ 0.8) screen found four exact TriviaQA collisions, two exact and two near Llama Questions collisions, and none in WebQ, SD-QA, or Reasoning-zh. One Llama match was in an unused generated-query file; seven rows were actually in the 5,228-row calibration merge. The reported 5,221-row refit changes internal AUC from 0.8761 to 0.8754 and, on separate official-configuration dumps, TriviaQA from 0.8113 to 0.8109 and Llama Questions from 0.7427 to 0.7411; reported remix accuracy shifts are below 0.5 point. The released exclusion list, refit artifact, and per-sample OOF scores now document the candidate; it was not substituted into the gate used for the main recorded arms. The above held-out AUC changes and remix effects remain log-reported, distinct from the OOF statistics checked above. They do not remove overlap from the presented benchmark, exhaust semantic or translated near-duplicates, or resolve base-model pretraining contamination. A revised deployment evaluation requires fresh arm runs with the deduplicated artifact and independently selected operating points.

A.1 Internal task strata

Table 3 describes all 240 original test queries under their original category assignments. These post-hoc summaries do not change membership, weights, thresholds, or the main result. The gate's gain over local answering varies substantially by task. SimpleQA traps show a large accuracy gain but consume expert calls on every query; that gain alone does not establish better selection than a random policy with the same call rate within that stratum.

Table 3: Descriptive strata of the fixed internal test set. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm's realized rate. The final column counts question waveforms longer than the expert's 30-second input window. One recorded run per query and arm; no samples are removed or reweighted.

Stratum	n	Local	A	Always	A call rate	$> 30\,\mathrm{s}$ (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The input-window limitation is concentrated in knowledge questions: 43 of 60 exceed 30 seconds, compared with three of 60 math questions. The always arm fires on 42 of those 43 knowledge queries, omitting their beginnings from its uplink transcription input; the remaining query is a non-commit. Knowledge accuracy remains 38.3% in that arm. This identifies an input constraint, without establishing how much of the remaining error truncation causes or predicting the gain from fixing it. All questions remain in the reported denominator.

The math stratum also illustrates why separate sampled arms cannot be treated as deterministic per-query counterfactuals. On its 43 GSM8K queries, local accuracy is 83.7% and aggressive accuracy is 74.4%, although the aggressive arm escalates none of them. This difference cannot be assigned to expert substitution. A single run per arm cannot separate sampling variability from differences in local-path execution conditions. The other 17 math queries are from MATH. We retain both subsets rather than selecting the one that improves the aggregate result.

A.2 Subsequent protocol and relay diagnostics

These follow-up artifacts are separate from the original pass in Table 1 and Figure 3; they do not replace its test mixture or operating points.

Previously completed internal repeats. The archived `repeat_variance.json` records two additional 240-query passes, r1 and r2, under the v1 protocol and original gate. Across r0/r1/r2, aggressive accuracy ranges from 62.9% to 65.4% (mean 64.4%, sample standard deviation 1.3 points); local accuracy ranges from 40.8% to 42.9%. These descriptive records include decoding and judge variability, rather than isolating either source. The main table retains r0 throughout, and the figure's query quantiles are not error bars from these repeats.

Validation excluded from the revised gate fit. The v2 protocol sends the expert only the audio prefix already consumed by the local model. It also synthesizes and saves output audio on both paths, transcribes that output for a separate content score, and records server monotonic timestamps. Its gate excludes 299 validation queries (239 English, 60 Mandarin) from the 5,221-row deduplicated merge, leaving 4,922 fit rows. Validation and the internal test set have disjoint query IDs. The recorded `val1` pass contains 299 queries per endpoint arm: local and always text accuracy are 152/299 (50.8%) and 231/299 (77.3%); output-audio transcription accuracy is 144/299 (48.2%) and 198/299 (66.2%). The latter measures transcribed content, not human-rated audio quality.

Using the never-arm onset scores and these cached endpoints, nominal 15/30/50% produces remix text accuracies of 57.5/66.2/71.6% at realized selection rates of 14.7/28.8/50.2%. These are validation remixes, not live executions of the three gate arms. The accompanying latency sweep adds chunk counts and component durations; it is still a reconstruction despite the availability of timestamps. It does not establish measured audio-ready latency or a latency-optimal tier choice. These validation observations therefore do not replace the main benchmark or establish the gain from any single protocol change.

Offline formatter checks. Applying formatter candidates v2 and v3 to the same archived expert texts recovers the two previously truncated numerical conclusions (64 and 36) with either candidate. For the known HTML-answer example, v2 returns an empty string and v3 preserves the answer content. Across 238 internal always-arm expert texts, empty outputs decrease from one to zero and a heuristic display-syntax flag decreases from 22 to eight; on 202 Mandarin reasoning texts, the flag decreases from nine to two. This effect is not uniform: SD-QA flags increase from two to three. These string checks neither measure TTS behavior nor establish higher answer accuracy. Similar reference-token retention cannot rule out other formatting errors, and inconsistent judgments of identical text prevent attributing every expert/relay score disagreement to the formatter. The new structured-answer candidate has no measured result here; its tool-free configuration also differs from the web-enabled expert baseline.

B Model roster

Table 4: Model roster. "Pair" groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

C Extended signal and readout analyses

Table 5: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix C.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	≈chance on raw backbones	—
p(True)-pre	0.807	trap 0.945 (probe: 0.232)	**collapses**
p(True)-post (full draft)	**0.899**	—	degrades
final-layer probe	0.822	**0.372** (inverts)	0.936 (intact)
mid-layer (L22) probe	0.866	**0.931**	**0.855** (text-calibrated)

The in-distribution shortcut, quantified. The numbers behind §3.1's claim that the final-layer probe's in-distribution AUC of 0.822 is mostly pool identity: a linear read of the same hidden state classifies a query's source pool at 95.8% accuracy; a pool-identity oracle alone reaches AUC 0.715; and appending true pool labels to the probe's features moves it 0.822→0.821. In-distribution area over random is statistically on par between the gate and the pool oracle (Appendix F); the two separate only under LOPO, where the probe inverts to 0.372 and the oracle is undefined.

ROC curves. Figure 4 shows the calibration-split ROC for the signals of Table 5.

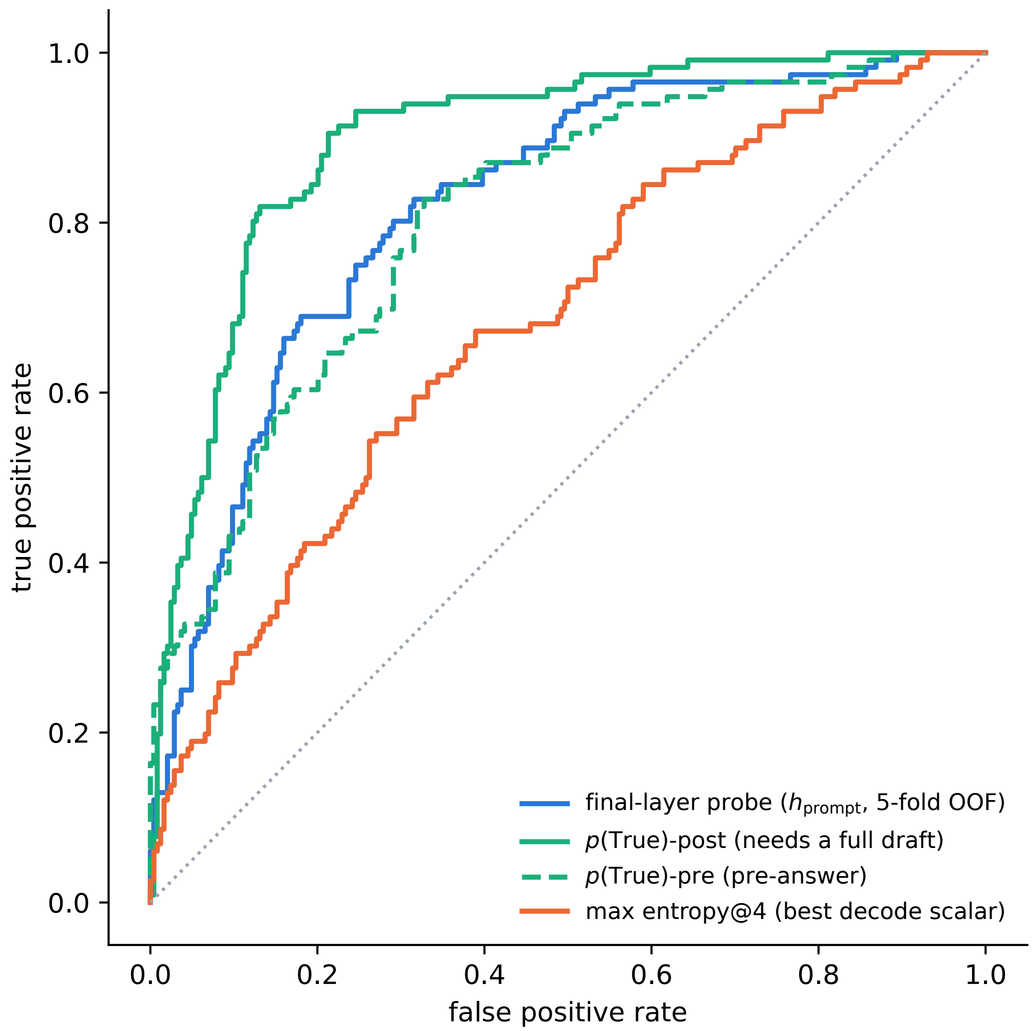

Figure 4: ROC on the calibration split. $p(\mathrm{True})$-post leads but needs a full drafted answer; the probes and $p(\mathrm{True})$-pre are pre-answer.

$p(\mathrm{True})$ backbone dependence. Across backbones, post-draft $p(\mathrm{True})$ beats the trained final-layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone's self-evaluation quality.

Matched pairs. Table 6: with label coverage matched by subsampling the raw control to the duplex model's per-pool fail rates, the raw backbone keeps LOPO math at 0.825/0.809 while duplex models sit at chance: label coverage is ruled out; the representation changed. Degradation is ordered raw $>$ omni-streaming $>$ duplex. Table 7 gives the by-depth summary behind Figure 2: the within-pair deficit at the best mid-layer is only -0.03 vs. -0.59 at the final layer; mean-pooled probes survive to the final layer on duplex models (LOPO math $\approx$0.80 at L35 on o4.5, 0.674 at L27 on o2.6, 0.13–0.15 below each model's mid-layer last-token peak, and uniformly weaker in-mix, -0.058 at L22); duplex damage is severe but local, omni-streaming damage mild but diffuse.

Modality specificity of the cliff. Under audio input the final layer does not invert (L35 audio LOPO math 0.936 vs. 0.366 text); a speak-mode template control on text input leaves the cliff unchanged (0.362): the operative variable is the modality of the input context, not the output mode.

Table 6: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	**0.809**	0.745	**0.538** $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	**0.526** $\approx$ chance	0.61

Table 7: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	**0.931 (L22/36)**	**0.366**	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	**0.822 (L21/28)**	**0.540**	cliff

Audio collapse of verbalized introspection. Table 8. Three controlled arms pin the mechanism on MiniCPM-o 4.5: not perception (ASR audit WER 0.074; $p(\mathrm{True})$ on the model's own transcript snaps back to 0.074; the well-heard subset still collapses); not a context prior (irrelevant filler audio does not inflate p_{yes}); binding (duplicating the question as text alongside the audio fully restores introspection, 0.034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at 0.93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking ("repeat-then-judge").

Table 8: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes}: text $\rightarrow$ audio	audio $p(\mathrm{True})$-pre AUC
MiniCPM-o 4.5	duplex	0.055 $\rightarrow$ **0.556** (collapse)	0.786 (trap dead)
MiniCPM-o 2.6	duplex	0.196 $\rightarrow$ 0.345	**0.491** $\approx$ chance
Qwen2.5-Omni	omni-streaming	0.279 $\rightarrow$ 0.213 (intact)	0.727
Freeze-Omni	frozen + adapter	0.165 $\rightarrow$ 0.112 (intact)	0.612 (capability caveat)

Probe $\oplus$ repeat-then-judge fusion. The restored self-evaluation is additive with the probe. We stack the deployed L22 probe with $p(\mathrm{True})$-pre asked over the model's own transcript (one short text prefill + a 1-token decode per turn; logistic stacker fit on the 360 calibration rows, native regime, Table 9). On the internal test split the fusion is the first significant AUC lift over the deployed probe (0.836→0.862; bootstrap 95% CI on Δ excludes zero), and it transfers to one external pool (TriviaQA 0.789→0.820); the remaining English pools move positively but within CI, and the Mandarin reasoning pool is untouched: there the self-evaluation itself carries no signal (AUC 0.586). Contrary to our a-priori expectation, the gain tracks where the self-evaluation signal is strong, not where the probe is weak; a diagnostic arm asking $p(\mathrm{True})$-pre on the original query text scores within 0.01–0.02 of the transcript arm on every pool, so transcription quality is not the limiter.

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (0.400→0.450), audio overconfidence (paired p_{yes} shift $+0.089$; failure subset 0.415→0.581), and the layer structure (text-calibrated read on real-speech audio holds 0.76–0.80 at L22–L26, peak 0.800 at L25, transfer across both content and modality).

Synthesis. End-to-end multimodal training builds the shared mid-layer core; duplex-style training additionally damages the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

The deployed probe, and which calibration each result uses. The mechanistic probe is a single-position L22 read (4096-d), fit on the 360-query calibration split; every representational result (§3,

Table 9: Probe $\oplus$ repeat-then-judge $p(\mathrm{True})$-pre fusion (native regime, deployed read point). Δ = fusion $-$ probe, bootstrap 95% CI.

pool	n	probe	rtj solo	fusion	Δ [95% CI]
internal test	240	0.836	0.836	**0.862**	+0.026 [+0.004, +0.048]
TriviaQA	250	0.789	0.761	**0.820**	+0.030 [+0.000, +0.062]
WebQ	250	0.745	0.723	0.771	+0.026 [−0.007, +0.059]
Llama-Q	250	0.686	0.674	0.691	+0.006 [−0.020, +0.031]
SD-QA	200	0.803	0.742	0.808	+0.004 [−0.023, +0.032]
Reasoning-zh	202	0.640	0.586	0.618	−0.021 [−0.076, +0.032]

Appendices C, E) is this probe. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean; 12,288-d), refit on a calibration mix widened to 2,310 public-benchmark queries sourced separately from every evaluation benchmark (overlap audit below); a pre-registered guard held (frozen internal test AUC 0.860→0.879) and feature selection never saw the externals. It drives the turn-based live loop (§5.1), the frozen transfer results (§5.2), and the talker-enabled replay, frozen throughout, calibrated on text and speech-rendered internal data only. The serving-regime validations are the exceptions. In the concurrent regime, the same 12,288-d feature set is refit in-regime on 360 rows for the loop of Table 16 and on the full 2,310-row mix to measure the effect of calibration scale (Appendix I). In the native regime, the same feature set is refit at the native listen→speak read point; the deployed native gate is the 5,228-row official-configuration merge of Table 20 (Appendix P). **Selection history.** L22 and the three-position feature set were chosen on internal calibration-split cross-validation before any external benchmark was scored (the 8-position $\times$ layer sweep and the 19-configuration feature sweep are both calibration-only), and the internal frozen test split was the pre-registered guard. The external pools were then read once with the frozen artifact. Two honest exposures remain: the layer band was picked on internal LOPO rather than by nested selection inside each held-out pool, and the second duplex family's mid-band peak (L30–34 of 56) was located with its own sweep, so both layer choices are best read as confirmed on held-out pools, not as untouched-holdout estimates. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe's in-distribution area advantage is modest: its real advantage is LOPO robustness, which an in-mix test cannot show.

D Second duplex family: NemotronLabs-VoiceChat-11B

Probe training protocol. Our latest probe uses the pass-3 replay through NVIDIA's native frame protocol with the VoiceChat checkpoint frozen. Of 2,550 English calibration queries, 2,481 produce a commit frame and enter the fit. At each of L26/L30/L34, we concatenate the commit state, the eighth post-commit state, the mean of the first eight post-commit states, and the running mean before commit (17,920 dimensions per layer). Each layer has a `StandardScaler`$\to$ L_2-logistic head ($C{=}10^{-4}$); we normalize the three training logits by their standard deviations and average them. A GPT-5.4-mini judge supplies pass/fail labels against reference answers. We select the recipe on calibration-only LOPO metrics, report five-fold stratified OOF AUC, and score the four external pools without fitting on their labels. Because the variant review repeatedly reported these external metrics, they are development evidence rather than untouched holdouts. The 15/30/50% thresholds are OOF-score quantiles. Because the feature window ends 640 ms after commit, this is an answer-onset read rather than a strictly causal pre-answer read.

Layer location. On the same expanded calibration cohort, the single-feature answer-onset sweep rises from 0.720 OOF AUC at L2 to 0.785 at L30, stays within 0.006 from L18 through L34, and falls to 0.761 at L54. Thus the readable band remains mid-network in this second model family; the reported L26/L30/L34 average spans its plateau.

Scope of the transfer test. The capture follows the model's native frame protocol, but it is offline: NeMo's released `offline_voicechat` path re-runs the full prefix every 80 ms and cannot sustain a real-time duplex loop. We therefore use this experiment only to test whether the probe method transfers to another model family and to external speech-QA pools. It does not measure real-time full-duplex serving, relay quality, or end-to-end latency.

Pass-3 results. The three-layer answer-onset probe reaches OOF AUC 0.818, pooled LOPO 0.792, and mean within-held-pool LOPO 0.691. With no external labels used for fitting, its AUC under the GPT-5.4-mini training judge is 0.827/0.862/0.790/0.775 on TriviaQA/WebQ/Llama-Q/SD-QA (mean 0.813; Table 10). Under the pool-aligned judges used for the cross-family comparison, the same scores give 0.845/0.845/0.803/0.770 (mean 0.816). At fixed 15/30/50% per-pool escalation budgets, four-pool mean accuracy is 0.550/0.650/0.757 versus 0.507/0.577/0.670 for matched-rate random. It beats random on every pool at 30% and 50% ($p \leq 0.0008$); SD-QA at 15% is inconclusive (p=0.063). On VoiceBench AlpacaEval, the same pass-3 ranking raises the 1–5 score from 3.55 locally to 3.89/4.17/4.51 at 15/30/50%, versus 3.76/3.96/4.23 for matched-rate random (n=198, all $p < .001$).

Read-point boundary. The reported feature window covers the first 640 ms after the model's commit, so it reads answer onset. A strictly causal probe that uses only the commit state, the preceding eight frames, and the running pre-commit mean loses about 0.03 external AUC (Figure 5). Replacing the full-user-audio mean with the online running mean changes external AUC by only -0.004 $[-0.009, +0.001]$. Fixed calibration thresholds still drift across pools: the latest probe fires on 3/9/20% of Llama Questions and 38/50/73% of SD-QA at nominal 15/30/50% budgets. Per-pool or windowed quantiles are therefore required for budget control.

Table 10: Pass-3 NVDA probe read points under GPT-5.4-mini labels (L_2-logistic C=10^{-4} on standardized features; 2,481-row analysis fits; external labels excluded from fitting, n = 249/250/249/200). LOPO$_{\text{pool}}$ is mean within-held-pool AUC over 13 eligible calibration pools. Δ is the paired-bootstrap difference in external mean from the single-layer answer-onset reference on the same replay.

read	features	OOF	LOPO$_{\text{pool}}$	external mean	Δ [95% CI]
strictly causal, L34	commit frame ‖ pre-mean$_8$ ‖ running mean	0.812	0.694	0.781	-0.029 $[-0.048, -0.009]$
strictly causal, L30	same	0.805	0.678	0.785	-0.025 $[-0.044, -0.006]$
single-layer answer onset, L30	onset-last ‖ onset-mean$_8$ ‖ user-mean	0.816	0.685	0.809	—
single-layer with running mean, L30	onset-last ‖ onset-mean$_8$ ‖ running mean	0.814	0.684	0.805	-0.004 $[-0.009, +0.001]$
reported 3-layer answer onset, L26/30/34	+ commit frame, running mean	0.818	0.691	0.813	$+0.004$ $[-0.005, +0.012]$

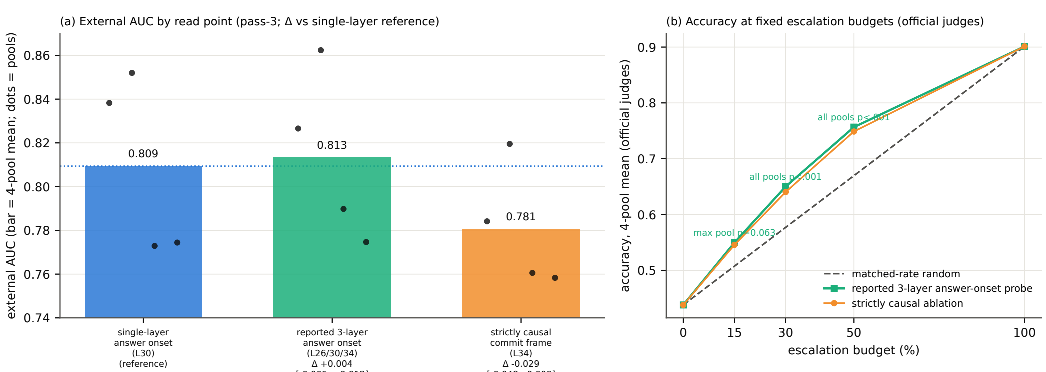

Figure 5: Pass-3 NVDA method-transfer results under native-frame replay. (a) External AUC under GPT-5.4-mini labels by read point: the reported three-layer answer-onset probe reaches 0.813 mean AUC, while removing post-commit states costs about 0.03. Dots are the four pools. (b) Accuracy at fixed escalation budgets under official judges: the reported probe beats matched-rate random at every budget on average.

E What breaks at the output: mechanism audit

§3.2 reports that the final-layer last-token read inverts. This section reports what we could and could not establish about why. Everything below uses the Phase-5d all-layer prefill dumps (every decoder layer, last-token and mean-pooled, same 600 queries) and the identical probe protocol as the layer sweep: calibration split, L_2 logistic regression, LOPO trained on the four non-math pools and scored on held-out math. The reproduction check recovers the published L22 and final-layer numbers to $\leq$0.012 (`scripts/14-17_*.py`).

Four mechanisms ruled out. Table 11 lists them. (i) **Speak mode.** Prefilling the same text queries under the TTS/speak-mode template leaves the cliff exactly where it was (final-layer LOPO math 0.362 vs. 0.366 plain), so the readout is not being displaced by a prepare-to-speak state. (ii) **Modality axis.** The mean audio-minus-text direction at the final layer is orthogonal to both the inverting final-layer probe and the transferring L22 probe (cosine 0.013 and 0.027; chance is $1/\sqrt{4096} = 0.016$), so the cliff is not the audio/text split that §3.3 measures. (iii) **More shortcut available late.** Source-pool identity is linearly decodable at 0.88–0.96 at every depth, in the duplex model and its raw backbone alike: the query-type shortcut of §3.1 is not something the late layers add; it is what remains once the competence signal stops being readable. (iv) **Wholesale representational drift.** Raw linear CKA between the duplex model and its backbone appears to collapse late (0.80→0.47), but transformer residual streams carry a few massive-activation coordinates that dominate the Gram matrix; with per-feature standardization the same comparison is nearly flat (0.82→0.78) and does not localize the cliff. Projecting out the massive-activation axes, or the pool-mean directions, rescues nothing (0.35–0.40 at every $k \leq 20$, against a random-direction control of 0.37).

Table 11: Mechanism candidates for the final-layer inversion. Final-layer LOPO hard-math AUC on MiniCPM-o 4.5 unless noted; 0.366 is the unmodified baseline and 0.931 the L22 read.

candidate	test	verdict
prepare-to-speak state	speak-mode template prefill	0.362, no change
modality (audio vs. text) axis	cosine with the probe direction	0.013 ≈ chance
late layers encode more pool identity	pool decoding by depth	0.88–0.96 at all depths
representation rewritten late	standardized CKA vs. backbone	0.82→0.78, flat
massive-activation coordinates	project out top-k such axes	0.35–0.40, no rescue
query-type directions	project out pool-mean directions	0.34–0.45, no rescue
delivery to the output position	**residual component of the read**	**0.93→0.27 (Fig. 6)**

What does separate the models. Figure 6 measures the probe rather than operating on the model. For the probe trained at each depth on the four non-math pools, we split its held-out score into the part carried by that layer's five dominant principal directions (estimated from training rows only) and the residual, $s = w^\top Px + w^\top (I - P)x$. The residual component is the distributed read, and it is the one that transfers: it holds 0.85–0.93 through mid-network on MiniCPM-o 4.5 and then decays to **0.27** at the output, with the same decay on MiniCPM-o 2.6 (0.81→0.52) and no decay on either raw backbone (Qwen3-8B 0.86, Qwen2.5-7B 0.71 at their final layers) or on the omni-streaming control (Qwen2.5-Omni holds 0.75 to its output with a flat variance share): the decay tracks duplex fine-tuning specifically, not streaming or audio fine-tuning in general. On MiniCPM-o 4.5 the score also concentrates: the dominant subspace carries 39% of the held-out score variance at the final layer against $\leq$0.18 at every earlier depth and 0.05 for its backbone; on MiniCPM-o 2.6 this concentration is present but weak (0.16 vs. 0.14 for its backbone), so we report it as a property of the stronger duplex model, not a general law.

Direct turn-control tests. Two experiments target the turn-control story head-on rather than by exclusion (`modal_interp.py`, `scripts/19_*.py`). **Control-token logit lens:** applying each model's own final norm and unembedding to the stored per-layer states, the probability mass on the trained-in control vocabulary (added/special tokens) rises over MiniCPM-o 4.5's last layers (log-mass -23.8 at L32 to -11.8 at the output) while the raw backbone's falls by fifteen log-units over the same span: the duplex output layer, uniquely, stays ready to emit control tokens. But the fine-grained prediction fails: the inverting probe direction is no more aligned with the control tokens' unembedding rows than with random vocabulary rows (max $|\cos|$ 0.069 vs. 0.068), and per-query control mass correlates with the probe score at only r=0.18. **Listen/speak intervention:** re-prefilling 60 calibration queries with a still-listening vs. answer-now suffix moves the final-layer state substantially in both models (relative displacement 0.27 duplex, 0.59 raw; cue text is just prompt semantics), and the cue direction is orthogonal to the inverting probe ($|\cos|$=0.019, chance 0.016), shifting its score by only $+0.23$ sd (and the deployed L22 read by $+0.08$ sd). Together: the duplex output layer is measurably control-biased in its output distribution, but the inversion is not carried by control-token directions or by an explicit listen/speak axis: the simple takeover story fails its own direct tests.

What we do not claim. Destructive tests do not settle the mechanism. Removing the top five principal directions from the final layer does raise LOPO math from 0.378 to 0.758, but the identical operation costs the raw backbone 0.90→0.14 and costs the intact L22 read 0.931→0.760, so it is a damaging intervention that happens to help a broken readout, not evidence of a duplex-added subspace. We therefore state the finding at the level the evidence supports: duplex fine-tuning does not erase the competence signal, it stops delivering it in distributed form to the final last-token position, and the layer where delivery fails is also the one whose output distribution stays control-biased. Attributing the loss to a specific duplex objective needs training-time access we do not have; the practical consequence, read mid-network, does not depend on the attribution.

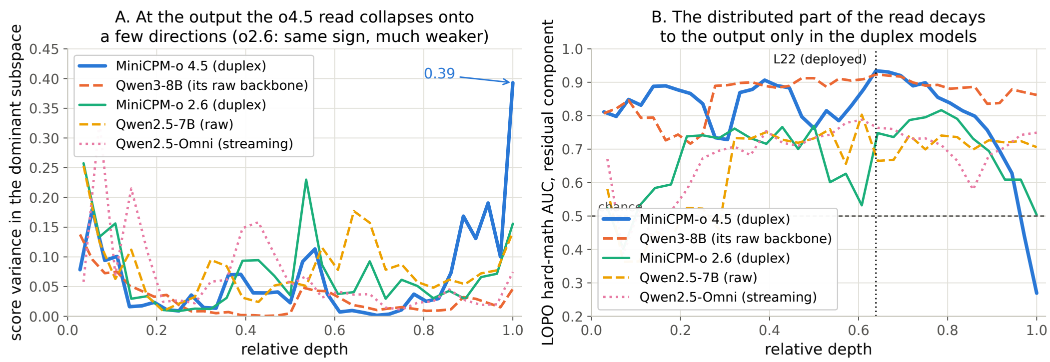

Figure 6: Non-destructive decomposition of the probe's held-out score at every depth. **Left:** share of score variance carried by the layer's five dominant directions. **Right:** transfer AUC of the residual (distributed) component, intact to the output on both raw backbones and on the omni-streaming control, decaying to inversion over the final layers of both duplex models.

F Query-surface router audit

A RouteLLM-style TF-IDF→LR router under identical labels reaches OOF AUC 0.669 (below the pool oracle's 0.715); a 24-configuration sweep tops out at 0.689, so this is the query surface's ceiling, not tuning. Under LOPO it collapses to 0.38–0.57 and scores the 100%-fail trap pool at 0.230: it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 8). In our RouterBench experiment [Hu et al., 2024], the identical recipe uses 36,497 prompts retained from the zero-shot release after dropping rows missing the prompt, benchmark identity, or either the Mixtral or GPT-4 score. This is our filtered subset, not the full benchmark size. The router reaches in-domain AUC 0.710 with leave-one-benchmark-out at chance; RouteLLM's released preference-trained routers score 0.523/0.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (0.440). On audio labels the router improves (test 0.814) but still trails the mid-layer probe (0.879) and cannot fire mid-stream. In distribution, the gate's area over random ($+0.054$) is statistically on par with the pool oracle's ($+0.042$; paired bootstrap delta CI $[-0.003, +0.027]$, p=0.13): the gate's case is transfer, not the in-distribution margin. $p(\mathrm{True})$ tradeoff variants: Figure 9.

G Gate timing details

Against the model's own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode, self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@33%: 78.7% at mean 6.5 s vs. all-THINK 63.7% at 22.4 s; Table 12). The reason is mechanistic: the gate flags knowledge and trap failures, which extra compute cannot fix, so external escalation is necessary, not merely better.

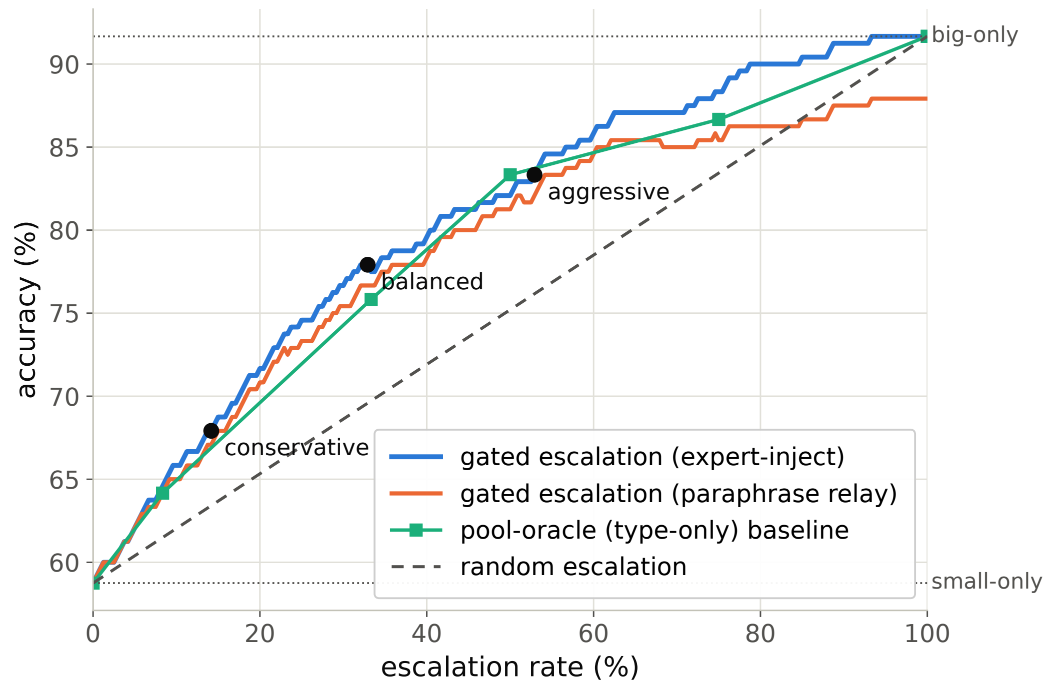

Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

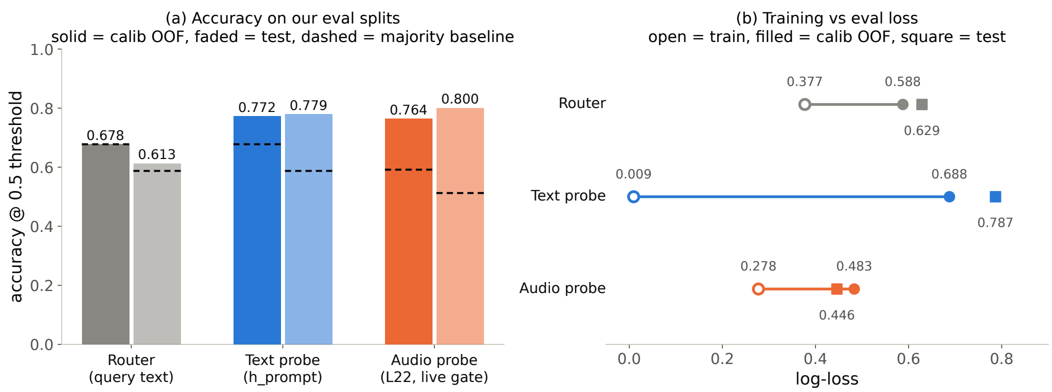

Figure 8: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

Measurement protocol. Gate latency is wall-clock from prefill start to the L22 score being available (truncated forward + probe dot product), CUDA-synchronized, one H100, 50 queries per modality, three warmup calls excluded, interleaved per query with the reference arms; P50/P95: gate 20/25 ms text and 45/104 ms audio vs. full-prefill TTFT 36/47 and 68/144 ms. Network and expert-RPC initiation are excluded: the cloud call fires asynchronously at the gate, and its user-facing cost is reported as expert-content-audible latency in Appendix H. Absolute prefill wall-times vary with call overhead across benches; the robust cross-bench statistic is the ratio: the L22 state is ready at ∼57–61% of prefill.

Per-layer truncated-forward cost vs. quality (Figure 10): the L22 hidden state is available at ∼57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at ∼55–60% depth. If the cloud call launches at the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but

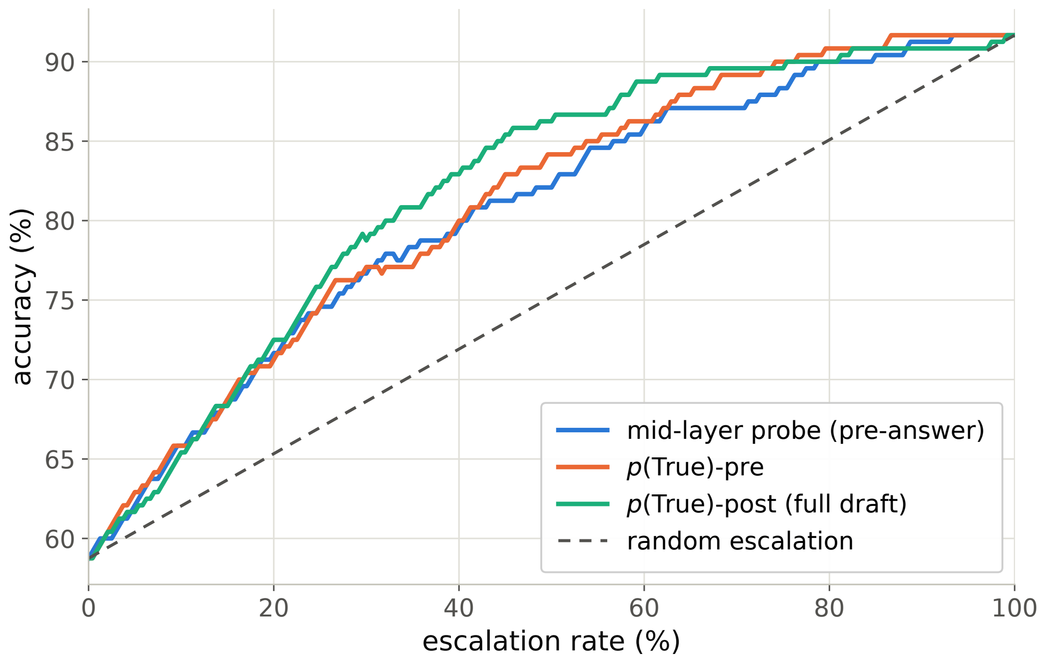

Figure 9: Offline tradeoff for the $p(\mathrm{True})$ signals (pre-answer and post-draft).

Table 12: Deployment policies, frozen test split (n=240); latencies measured.

policy	acc (%)	lat. mean (s)	P50	P95
fast-only	58.8	3.0	3.5	4.2
all-THINK (built-in)	63.7	22.4	17.2	60.1
gated-think @33%	61.3	12.1	3.6	47.6
gated-cloud @15%	68.8	5.3	3.6	10.1
gated-cloud @33%	**78.7**	**6.5**	3.6	20.8
gated-cloud @50%	85.8	7.0	3.6	24.4

only ∼20–31% of escalated queries (they skew to short local answers); the residual wait is P50 2–3 s, P90 ∼27 s (Figure 11), the stall-phrase budget the injection protocol must cover.

H Live-loop details

What the signal detects. The pre-answer probe is specifically a retrieval-failure detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is observable in the model's state. Execution failures (stuck mid-derivation) surface in decode-time signals instead, and metacognitive failures (false-premise questions, where no failure event occurs at all) are invisible to every signal we tested, query-surface routers included (escalation rate 18% vs. trap's 90%; Appendix H), which motivates a decode-time check behind the pre-answer gate.

Benchmark and historical tables. The upper block of Table 13 and Table 14 summarize native answer-content accuracy and reconstructed time to first audio. The lower block and the audible-onset experiment below concern the retired turn-based harness. Their timing endpoints and thresholds differ (Appendix A). In the native table, conservative leaves P95 essentially unchanged (17.3 to 17.5 s) but raises P99 to 80 s; the earlier turn-based tail reduction does not carry over to these rows. Figure 13 illustrates historical recorded timers.

Historical turn-based time to first audible audio. The earlier sweep is text-out; a TTS-on re-run of the balanced arm (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker's own voice) measures speech onset directly. The gate decision is unchanged: the TTS token

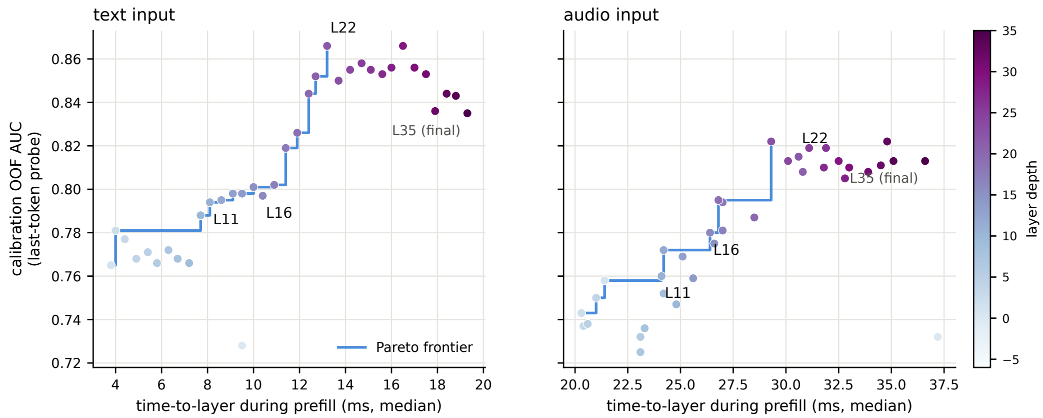

Figure 10: Fork-depth Pareto: per-layer wall-time vs. gate quality.

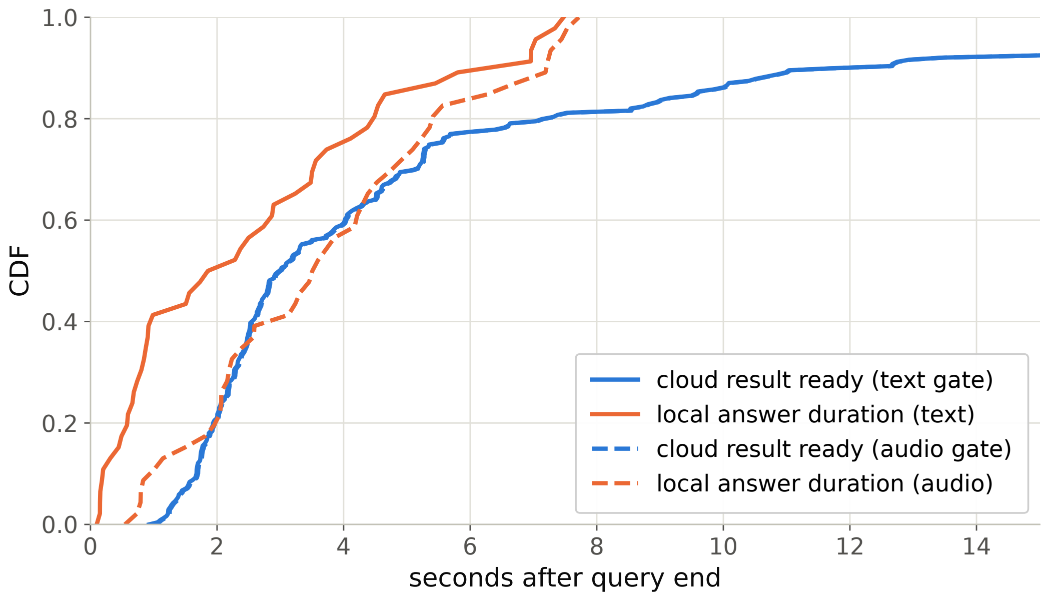

Figure 11: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

enters the context only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From gate decision to first audible audio: **local answers p50 0.552 s / p95 0.599 / p99 0.644**; **escalated turns p50 0.022 s / p99 0.029**: the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the escalated path reaches first audio 25× faster. Expert content becomes audible at p50 6.4 s / p95 26.0 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 0.66 s after the answer returns); per-turn dead air between stall end and expert content is p50 2.5 s / p95 22.4 / p99 30.7, with 25/84 escalated turns fully covered; the stall covers the head of that wait, and Table 14's completion profile bounds the tail.

Speculative prefetch, measured and dropped. Framed as speculative escalation (draft–verify at sentence granularity), its acceptance rate (does an expert answer to the partial transcript answer the full question) is 0.07/0.19/0.51 at 25/50/75% of the audio; at the mid-stream gate's typical fire point 6–8 of 10 calls would be wasted. The trap pool's floor (0.00/0.10/0.27) shares the back-loaded-core property that breaks mid-stream probe reads.

Table 13: Native-protocol answer-content sweep on the internal test set (n=240), query-bootstrap 95% CIs (B=5000); one recorded run per query and tier. Random follows the original Table 1 mixture convention, using the gate rate and the always-policy endpoint. Expert returned text scores 0.765 on 238 rows. Top: native sessions with the TTS relay. Bottom, for reference: the retired turn-based harness loop on the same queries (audio input, text output).

tier	esc	content	Δ vs. floor	random
never	0%	0.408 [0.35,0.47]	—	0.408
conservative	8%	0.467 [0.40,0.53]	+0.058	0.433
balanced	25%	0.542 [0.48,0.60]	+0.133	0.482
aggressive	52%	0.629 [0.57,0.69]	+0.221	0.561
always (measured)	99.2%	0.704 [0.65,0.76]	+0.296	0.704
turn-based harness loop (retired), full pool				
never	0%	0.383 [0.32,0.45]	—	0.383
conservative	16%	0.483 [0.42,0.55]	+0.100	0.420
balanced	35%	0.554 [0.49,0.62]	+0.171	0.465
aggressive	61%	0.596 [0.53,0.66]	+0.212	0.526
always (measured)	100%	0.617	+0.234	0.617

Table 14: Reconstructed time to first audio (TTFA, s), internal native benchmark, n=240 per arm. Escalated rows end when the relay waveform has been synthesized (expert wait plus blocking TTS); local rows end at answer-text completion, an upper bound on their first audio. The relay waveform's own playback duration ("play", mean over escalated rows) is not latency and is listed separately; it is inflated by code, LaTeX, and markdown in expert answers being synthesized verbatim. These are not client-side audible completion times (Appendix A). P95/P99 summarize the sampled query distribution and are unstable at this sample size.

tier	esc	mean	P50	P95	P99	ΔP50	play
never	0%	5.7	3.6	17.3	21.2	—	—
conservative	8%	7.6	4.2	17.5	80.2	+0.6	53.2
balanced	25%	8.7	5.3	22.9	89.6	+1.7	26.4
aggressive	52%	14.4	8.0	60.8	131	+4.4	25.0
always	99.2%	17.6	11.5	69.4	117	+7.9	27.8

Conflicting-injection controls (172 sessions). Fabricated conflicting expert answers by within-pool derangement, four framings: worst-case comply is 0.00 neutral / 0.53 deployed authority / 0.79 strong-override, while seeding words into the talker's mouth backfires (comply 0.25, lip-service 0.62; forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed. Model-wrong × wrong-injection swallow rate is 0.64: expert quality is the accuracy ceiling of the cascade.

Uplink diagnostics. On the escalated population the expert answering the talker's own transcript scores 0.585 vs. 0.871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (0.585→0.694, McNemar p=0.007), diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the audio channel itself is lossless for speakable content: the leak is the talker's transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated failures, first-generation sweep).

Boundary query classes. **False premises**: the model fails substantially (0.63 text / 0.47 audio) and the expert would help (0.80), but the gate is blind (fire@balanced 0.18 vs. trap's 0.90; no score separation): answering along a false premise does not feel like inability. Real-time queries ("what is NVDA trading at"): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing escalation rates to 0.80/0.93/1.0 across tiers with frozen-test AUC unchanged (0.877 vs. 0.879). One calibration subtlety: tier budgets

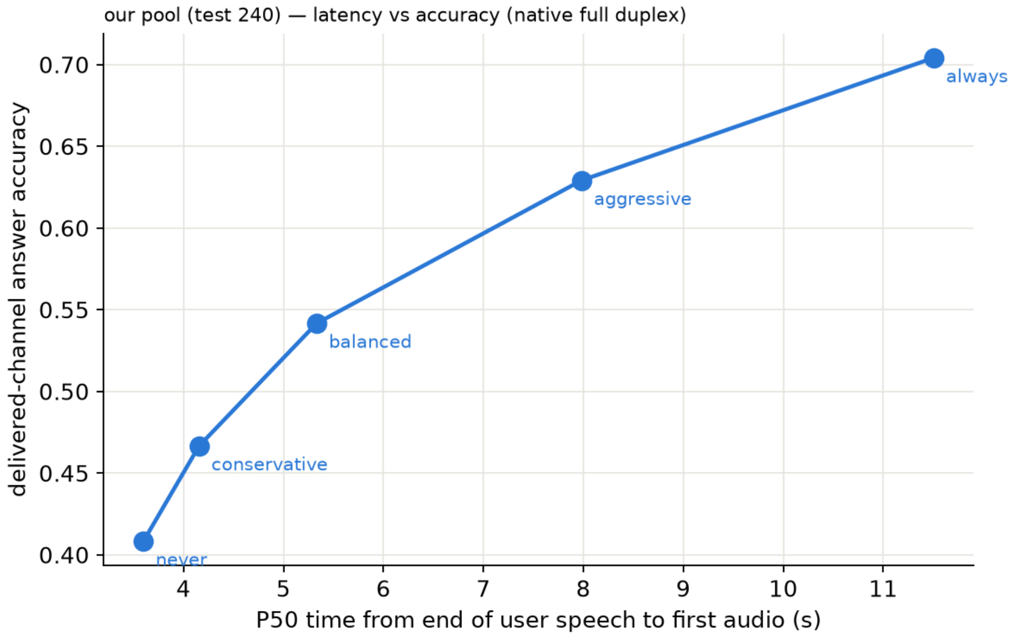

Figure 12: Historical plotting view of native answer-content accuracy against reconstructed P50 time to first audio; see the endpoint caveats in Appendix A.

must remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Demonstration system. The pipeline runs as an interactive web demo on one H100, on the model's native full-duplex head (Appendix P): every second of microphone audio is prefilled into the context the talker is generating from, the head itself decides listen/speak per chunk, and the gate reads L22 at the listen→speak transition against the in-regime thresholds. Escalated turns play a canned stall (teacher-forced once, the talker's own voice), consult the expert (with a demo-only web-search tool for real-time queries), and the verified answer is spoken verbatim in the talker's voice, paced on the duplex chunk cadence; whatever was actually played—the stall line, the answer up to any interruption—is teacher-forced into the head's context as its own completed speech turn, so the context the head takes its next listen/speak decision from matches what the user heard, the same shape a native turn leaves. The delivery is an ordinary duplex turn, hence interruptible like any other. There is no VAD, no ASR barge-in classifier, and no abort machinery: floor control is entirely the duplex head's own behavior (Appendix P). An earlier demo build approximated barge-in with an energy-VAD + burst-ASR harness over the turn-based loop; it is retired, and survives only as the harness-overlap arm measured above. The demo will be made available upon publication.

I Duplex-validation sweep: the gate under the talker

The talker-disabled live sweep (§5.1) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean**: the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS 0.008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap**: an easy warmup question (probe score <0.25, locally answered, >5 s spoken answer) puts the talker mid-answer, and the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Figure 13: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: talker-disabled AUC 0.866 $[0.817, 0.909]$, clean 0.871 $[0.824, 0.914]$, overlap 0.856 $[0.803, 0.901]$; barged-only subset (n=237) 0.854. Paired bootstrap deltas against the talker-disabled arm are $+0.005$ $[-0.025, +0.035]$ (clean) and -0.009 $[-0.036, +0.016]$ (overlap), both inside the ± 0.02–0.03 live replication floor. Per-query scores correlate across regimes at r=0.90–0.92, and the balanced threshold makes the same escalate-or-hold decision as the talker-disabled read on 0.887 (clean) and 0.912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 (p95 $\leq$30 ms) in both arms, with the first audible answer chunk arriving $\geq$0.5 s after the gate decision (p50 $\approx$0.58 s), so the decision precedes any audible commitment. Scope: this loop keeps per-turn session control: the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled during generation. In hindsight this sweep and the concurrent arm are best read as read-point ablations: the deployed native full-duplex regime (Appendix P) sits strictly between the turn-based ideal and the harness-concurrent worst case on every pool measured. The concurrent regime is measured separately below.

Floor control under escalation: barge-in vs. backchannel. The demonstration loop's floor policy (speech over the talker ducks playback; a fast ASR pass returns the floor on backchannels or commits the interrupt) raises a distinct worry: did adding the gate change the talker's overlap behavior? A dedicated sweep injects overlap stimuli into the deployed loop at real-time pacing, four classes: short backchannels (0.40–0.46 s, the English+Chinese lexicon), long lexicon-only continuers (time-stretched to 1.33–2.08 s, probing the sustained-speech commit rule), out-of-lexicon stop commands, and pool queries spoken over the answer, in matched pairs with the gate on and off under an identical floor policy (non-firing queries, n=40 per class per arm). The answer is exact orthogonality: all 160 pairs make the identical duck/resume/interrupt decision in both arms (every paired rate delta 0.000 $[0.000, 0.000]$; paired latency medians within 31 ms), down to the single seeding failure, which occurs on the same query in both arms. By construction the gate reads once at end of turn and never enters the floor state machine; this measures it. The three phases that exist only under escalation (the stall sentence, the expert wait, the relay speech) are the actual new risk surface and behave (Table 15): a genuine barge-in aborts the escalated turn and seeds the next turn 40/40, and short backchannels hold the floor at the same rate as during local answers. One asymmetry is disclosed: a false interrupt during the expert wait cancels the pending expert call, so the policy's fail-toward-interrupt bias is recoverable but not free there. That residual backchannel false-interrupt rate (0.40 short, 0.50 long; identical in both arms) is a property of the harness policy and is per-stimulus deterministic: non-lexical hums ("mm-hm" and its Chinese counterpart) die by the ASR-empty fail-closed rule (16/16 each; worded tokens such as "okay" 0/16), and continuers sustaining loud speech past the $\geq$1.2 s commit rule die without any ASR check (20/20; pause-bearing variants 0/20). We keep the policy as deployed for measurement integrity and note both rules as the obvious fixes. For scope: the checkpoint's native duplex profile, measured separately in our single-pass evaluation on Full-Duplex-Bench v1.0 [Lin et al., 2025] (727 samples, official scripts), shows lower pause turn-over rates than the baselines listed in the benchmark paper (0.125/0.117 Candor/synthetic), a user-interruption turn-over rate of 0.915 with 0.90 s latency, and zero spontaneous backchannels. Our evaluation uses Parakeet ASR, which differs from the original paper's ASR setup; the historical baseline comparison is therefore not a matched rerun. These MiniCPM-o 4.5 measurements are ours, not results reported by Lin et al. [2025]. The lexicon floor is a harness-level approximation added precisely because the talker head has no native backchannel behavior to preserve.

overlap injected during	backchannel holds floor	barge-in aborts	seeds next turn
local answer (gate off)	0.60 [0.45, 0.75]	40/40	39/40
local answer (gate on)	0.60 [0.45, 0.75]	40/40	39/40
stall sentence	0.63 [0.38, 0.88]	16/16	16/16
expert wait	0.60 [0.33, 0.87]	14/14	14/14
relay speech	0.55 [0.27, 0.82]	10/10	10/10

Table 15: Floor-control sweep over the deployed voice loop. Short backchannels (lexicon bursts) vs. genuine barge-ins (pool queries spoken over the talker), by the phase the overlap lands in; the last three phases exist only under escalation. Interrupt latency p50 1.39–1.40 s from overlap onset in every phase; escalated-turn aborts discard the pending expert answer and the barge-in speech becomes the next turn.

Concurrent prefill: interleaved listen/speak in one KV stream. A third arm drives the streaming API in time division: while the talker is mid-answer, each 1 s chunk of the target query is prefilled into the same session between generation chunks, so the end-of-turn read fires while generation is still active (238/240 queries, full interleave coverage). The carrier answer is teacher-forced: under sampling the duplex model reliably yields the turn: it emits EOS within $\sim$2 s of user audio appearing in its context, the trained turn-taking behavior surfacing under interleave. This is the one shift on the transfer ladder that breaks the frozen read: scores drift up (mean 0.48$\rightarrow$0.71; the frozen balanced threshold's escalation rate jumps 37$\rightarrow$75%), AUC drops 0.869$\rightarrow$0.758 [0.693, 0.816] (paired ΔAUC -0.109 [-0.158, -0.064]), and label-free requantiling recovers decision agreement only to 75%. The information, however, survives in the state: refitting the same 12288-d linear read on 360 calibration queries collected in-regime reaches **0.818** [0.761, 0.871] on the held-out 240 (calibration OOF 0.775), more than half the gap recovered from those 360 in-regime rows alone. The mid-layer competence direction persists under concurrent listen/speak; unlike every earlier shift (modality, external pools, talker enabled), calibration must follow the regime. Run-to-run replication of the concurrent read correlates at 0.895. Caveat: the interleave goes through the public streaming API: turn-role headers enter mid-generation and the carrier text is forced, so this measures the concurrent state, not the official duplex serving format.

The full loop in the concurrent regime. With the in-regime gate (thresholds = label-free quantiles of each pool's own concurrent always-local arm scores), the complete escalation loop (carrier speaking, target audio interleaved, EOT read mid-generation, stall $\rightarrow$ expert $\rightarrow$ relay) was re-run end to end on all seven pools (Table 16). We report this sweep as exploratory: one session per query per arm, an in-regime probe calibrated on 360 internal rows only, teacher-forced carrier (the caveat above). The loop's mechanics reproduce: realized escalation rates track nominal on the external pools (10.8–51.2%), and latency holds (EOT read p50 23 ms, canned-stall onset 27–34 ms, local first audio p50 .655 s vs. .552 turn-based, the price of the longer concurrent context), and on the internal pool, where calibration matches the regime, so does selection: accuracy climbs 40.4$\rightarrow$52.1$\rightarrow$66.2% (speakable

44.5→56.0→71.1%) over the balanced and aggressive tiers, clearing the matched-rate random remix of the same never/always outcomes (p=.017 and 5×10^{-5}; speakable .009 and 5×10^{-5}), and selective escalation beats always-escalate (66.2 vs. 61.7% at two thirds of the calls; speakable 71.1 vs. 66.5): the turn-based signature result carries over. The matched-random rows of Table 16 bound any stronger reading, in three ways. (i) Gains are not monotone everywhere: the internal conservative tier under-fires (2.5% realized vs. 15.8% nominal; the top calibration quantile did not transfer to the regime; balanced and aggressive did) and lands flat (40.4→40.0, 44.5→43.6, inside the ±2–3 floor). (ii) Externally the 360-row in-regime probe separates from matched-rate random only in spots (Llama @50%: 90.8 vs. 86.5, p=3×10^{-4}; Reasoning zh @30/50%: $p \leq .0024$; AlpacaEval @15%: p=.031); at the other external operating points the budget gains are what random escalation would buy, consistent with this probe's external AUC of .57–.67 against the turn-based probe's .68–.81 (internally, where calibration matches the distribution, it reads .857 vs. the turn-based .879). We then tested the calibration-scale attribution directly: the full 2,310-row calibration mix was re-collected in the concurrent state and the same read refit at full scale. Scale is real: every English pool lifts (paired against the 360-row probe on the same features: TriviaQA .645→.699, WebQ .641→.713, Llama Q. .674→.755, SD-QA .639→.701; mean ΔAUC +.068 [.036, .099]) along a monotone data-scaling curve (external mean .625→.689 over 360→2,310 rows; internal .818 unchanged), but it buys only about a third of the distance to the turn-based .785–.806, and Reasoning zh, absent from the English calibration mix, moves within noise (.615→.577 [−.117, .037]). The residue is the regime itself, not calibration scale. Table 16's loop ran with the 360-row probe and its numbers are unchanged. (iii) On Llama Questions selective-vs-always is judge- and noise-sensitive (official judge 90.8 vs. 91.2; ours 86.4 vs. 85.6; both deltas inside the floor), so we claim that comparison only on the internal pool. Floors move by pool, the regime cost story: Llama ≈ unchanged, TriviaQA −16, Reasoning zh −18 (the English carrier hurts the zh pool most), WebQ +4.8 under the official judge (within the ±2–3 replication floor; the larger our-judge shift is alias-matching style sensitivity). AlpacaEval rises 4.36→4.60 (1–5 scale, always 4.76), but only its conservative tier clears matched random (p=.031; 4.40–4.55 across tiers), so the turn-based honest-negative stands.

Table 16: The full escalation loop in the concurrent regime (exploratory: one session per arm, 360-row in-regime calibration, teacher-forced carrier), all pools, judge-matched to the MiniCPM upper block of Table 1 (OAB official for TriviaQA/WebQ/Llama; ours for SD-QA/Reasoning/internal; VoiceBench 1–5 for AlpacaEval). Gate = in-regime probe, per-pool label-free quantile thresholds; tier columns are headed by each pool's realized rate, and the *rand* row under each pool is the matched-rate random remix of the same never/always outcomes (20k draws; p = P(random ≥ gate), marked $^{*}p<.05$, $^{**}p<.01$). Accuracies in %; AUC = in-regime probe vs. always-local arm failure, decimals. Script: `scripts/20_conclive_random_check.py`.

pool	never	cons.	bal.	aggr.	always	AUC
TriviaQA	50.4	54.0	64.0	71.6	91.6	0.636
rand (10.8/30.8/46.8%)		54.9	63.1	69.6		
WebQ	61.6	65.2	68.8	71.2	81.6	0.636
rand (15.6/30.4/48.8%)		64.7	67.7	71.4		
Llama Q.	81.6	83.2	84.8	90.8**	91.2	0.672
rand (14.8/29.2/51.2%)		83.0	84.4	86.5		
SD-QA	49.5	55.5	60.5	70.5	89.5	0.636
rand (15.0/29.5/50.5%)		55.5	61.3	69.7		
Reason. zh	40.6	48.5	58.4**	64.4**	80.7	0.570
rand (15.3/30.2/45.0%)		46.8	52.7	58.7		
internal (240)	40.4	40.0	52.1*	66.2**	61.7	—
rand (2.5/36.2/66.2%)		41.0	48.1	54.5		
internal speakable (218)	44.5	43.6	56.0**	**71.1****	66.5	0.857
rand (2.3/31.2/62.8%)		45.0	51.4	58.3		
AlpacaEval (1–5)	4.36	4.45*	4.53	4.60	4.76	—
rand (10.1/30.7/47.7%)		4.40	4.48	4.55		

A serendipitous recovery behavior. An accidental run surfaced one bonus behavior worth reporting: when a long question contains a >1.25 s internal pause, the endpointer fires early and the talker starts answering the fragment, but the question's own continuation then triggers barge-in, and the remainder becomes the next turn, where the gate simply re-scores it. Early end-pointing thus degrades into an extra turn rather than a lost query, the honest duplex-ish recovery available to a turn-based loop.

J Fixed-threshold transfer and the expert-benefit oracle

Both analyses below concern the retired turn-based experiment and use the gold-inject view (escalated rows take the expert's answer to the gold text). In that experiment, the most permissive recorded routing arm escalated 50%, while separate expert-reference runs covered every ID for the counterfactual mixtures. Labels use each pool's assigned judge; the main MiniCPM accuracy block uses the same judge families, but a different serving protocol. The AUCs in this appendix section come from the turn-based sweep's stored end-of-turn scores. They differ from a separate turn-based scoring pass by up to 0.03 externally and 0.04 internally (0.840 versus 0.879). Those passes render and prefill audio independently. These are failure-prediction AUCs; they are not columns of Table 1, which reports answer accuracy. AlpacaEval is excluded: it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe's ranking carries from what its scale carries, we freeze the single absolute threshold the internal system actually deployed (τ=0.628, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer**: the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does**: at whatever rate it fires, fixed-τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 17). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against local failure, but what a routing system needs is expert benefit: y=1 when the local model is wrong and the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA 0.796→0.782, Llama Questions 0.830→0.813) and more where the expert is itself weak (Web Questions 0.773→0.689 against a 0.864 expert ceiling; internal 0.840→0.732). The gate thus remains a usable benefit predictor out of distribution (0.63–0.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate's top-scoring 15% are, and that share climbs monotonically as the cut widens (0.655/0.685/0.711 at 15/30/50%): the gate's most confident failure calls are the ones the expert also misses. A benefit-trained refit closes none of this: with expert outcomes collected for all 2,310 calibration queries (gold-text expert, expert-right rate .853), retraining the same 12,288-d read directly on the benefit label moves internal benefit-AUC .732→.759 (paired $+$.027 [.004, .052]) and leaves every external pool inside noise (Δ $-$.029 to $+$.037, all CIs spanning zero), at no internal cost on the fail label (.840→.839). The benefit gap reflects the label's dependence on the expert's own failure surface, not a trainable objective mismatch: the expert-agnostic label leaves nothing on the table. Table 18 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

K Native escalation rates and historical transfer latency

Rates for the main MiniCPM block. Table 19 supplies the per-pool realized rates referenced by the caption of Table 1. They differ from the nominal 15/30/50% calibration budgets. The NVDA lower block uses its own 15/30/50% replay budgets and is not covered by this rate table.

Scope of the historical analysis below. The remaining paragraphs describe the retired turn-based transfer experiment. Its thresholds, local floors, and latency endpoints differ from the MiniCPM native benchmark; those numerical claims should not be applied to the main-table native arms.

Table 17: One absolute threshold (τ=0.628, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed-τ accuracy minus a random reference at the same realized rate (paired bootstrap over ids, B=10^4, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit (y= local wrong $\wedge$ expert right, base rate in the last column). Rates and accuracies in %; AUC in decimals.

pool	n	rate	local	acc	rand	Δ 95% CI	AUC_{fail}	$AUC_{benefit}$	base
Speech TriviaQA	250	28.4	66.4	83.2	75.0	[+4.4, +12.1]	0.796	0.782	30.8
Speech Web Questions	250	36.0	56.8	72.0	67.5	[+0.8, +8.4]	0.773	0.689	32.4
Llama Questions	250	8.0	83.6	86.8	84.3	[+0.6, +4.6]	0.830	0.813	11.2
SD-QA	200	48.0	51.5	80.5	71.4	[+4.5, +13.6]	0.800	0.765	43.5
Reasoning QA (zh)	202	2.0	58.9	60.9	59.5	[−0.1, +3.5]	0.679	0.634	30.2
internal frozen test	240	35.0	38.3	65.4	57.0	[+3.8, +13.1]	0.840	0.732	53.3

Table 18: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true y=1 cases first. The gate clears random everywhere; the oracle column is the remaining headroom. Accuracies in %.

	15% budget			30% budget			50% budget		
pool	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	71.0	76.0	81.6	75.5	83.6	96.4	81.6	91.2	97.2
Speech Web Questions	61.2	64.0	72.0	65.7	70.0	86.8	71.6	76.4	89.2
Llama Questions	85.0	88.8	94.8	86.4	91.2	94.8	88.2	93.2	94.8
SD-QA	57.7	63.5	66.5	63.9	74.5	81.5	72.2	82.0	95.0
Reasoning QA (zh)	63.1	67.8	73.8	67.4	72.8	89.1	73.0	76.7	89.1
internal frozen test	46.3	49.2	53.3	54.3	60.8	68.3	65.0	74.2	88.3

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail 0.16) the 50% cap is 0.32 and the deployed probe reaches 0.304, 95% of the cap (recall 0.93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose local decode is longest: on retrieval pools wrong answers hedge at length (decode time and answer length r=0.97), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: always-local arm local P50 1.711 s vs. 1.198 s kept; local accuracy 0.32 vs. 0.73). Controls: the probe does not key on length (corr(score, length) = $+0.05$; $\approx$0 conditioned on wrongness), and on SD-QA, where wrong answers are short, the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-local P50 falls 1.52→0.43 s across tiers (Figure 14). On Reasoning QA the shape inverts usefully: the local chain-of-thought is spoken, so every reasoning token enters the response clock, while the expert's chain is hidden behind a flat ∼3.7 s round-trip: escalation truncates the latency tail (P90 13.25→10.94 s) rather than fattening it.

AlpacaEval scale caveats. The official 4.81 [Cui et al., 2026, Table 4], rounded elsewhere to 4.8, is an offline chat-mode number (our chat-mode control reproduces it at 4.86) and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop's spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers; random picks would harvest the same gain.

L Full-pool and per-family figures

Figures 15 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix M); the noise audit behind the ±0.02–0.03 floor is Figure 16.

Table 19: Realized escalation rates (%) for the MiniCPM native results in the upper block and caption of Table 1. These are measured rates, distinct from the nominal calibration budgets and from the NVDA replay budgets. Local-only makes no expert calls.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

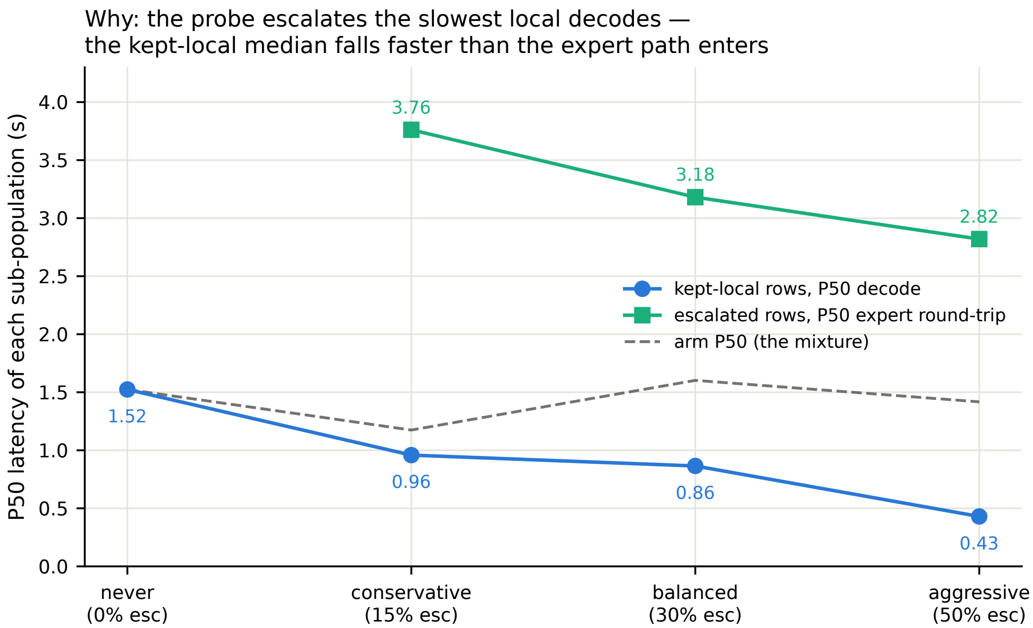

Figure 14: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ∼3 s expert round-trip; the arm median (dashed) is their mixture.

M External-benchmark figure set

The following are retained historical plotting views of the native benchmark. Their "delivered" labels refer to answer content, and "time-to-answer" axes show the reconstructed time to first audio defined in Appendix A. Expert-text lines are separate reference scores, not guaranteed bounds on the deployed channel; the updated main plot aligns the horizontal reference with the measured always arm and follows the original main-table random mixture convention.

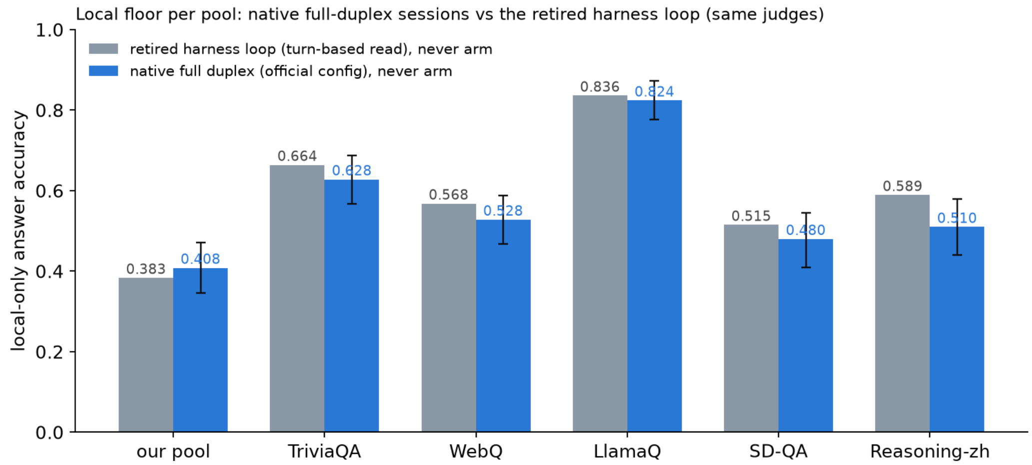

Figure 15: Local (never-arm) accuracy per pool: native full-duplex sessions vs. the retired turn-based harness loop, same judges. The native floor is 1–8 points lower on five pools and 2.5 points higher on the internal set; bars are bootstrap 95% CIs.

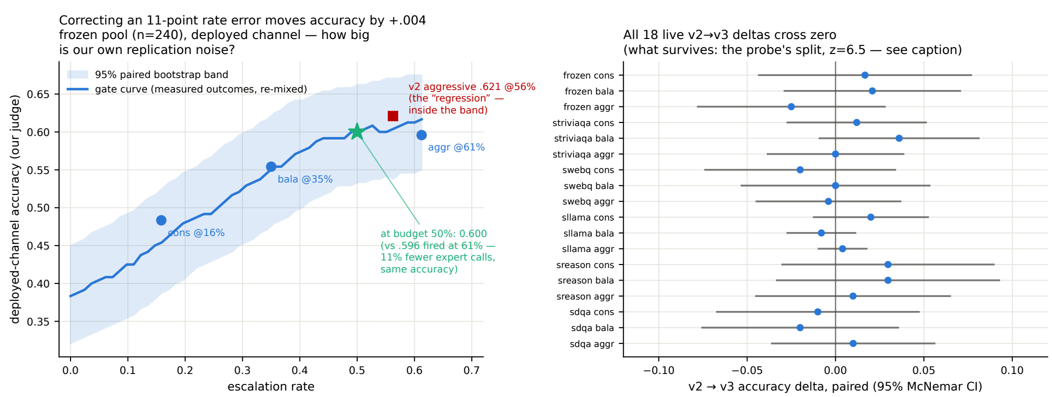

Figure 16: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE 0.009–0.028.

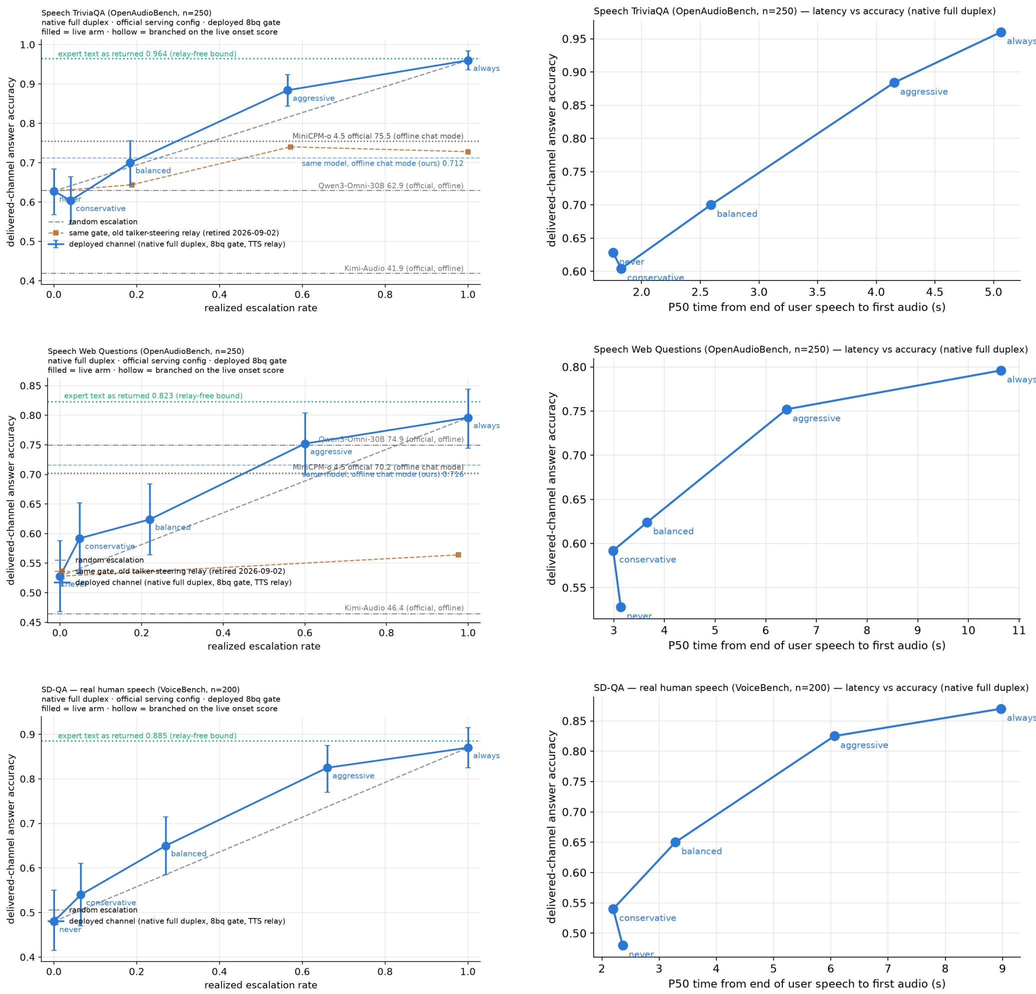

Figure 17: Speech TriviaQA, Speech Web Questions, SD-QA: native full-duplex live arms, delivered accuracy vs. realized escalation rate (left) and vs. P50 time-to-answer (right). Orange dashed on TriviaQA and WebQ: the same gate with the retired talker-steering relay.

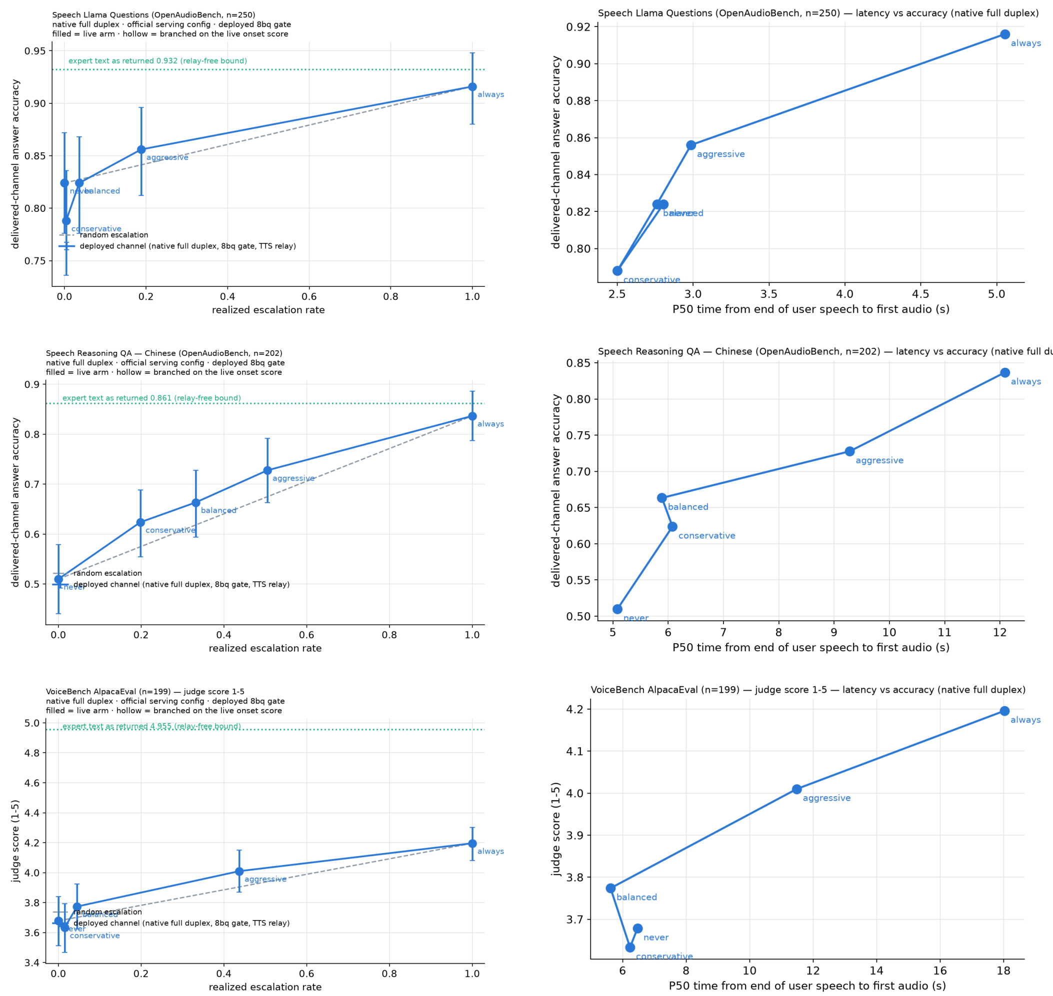

Figure 18: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: native full-duplex live arms, delivered accuracy (AlpacaEval: judge score, 1–5) vs. realized escalation rate (left) and vs. P50 time-to-answer (right).

N Judge and self-evaluation prompts

All adequacy labels use the following fixed system prompt (structured outputs with a strict {`reason`, `adequate`} schema; the judge never sees which system produced the answer):

```
You are a strict, fair grader of a small language model's answers.
You are given a QUESTION, an optional REFERENCE answer, and a
CANDIDATE answer produced by a small model.  Decide whether the
CANDIDATE is ADEQUATE, meaning ALL of:  (1) factually correct;
(2) any reasoning shown is valid; (3) it actually and completely
answers the question that was asked.  For multiple-choice questions
the candidate must land on the correct option (by letter or by
unambiguously stating the option's content).  A partially correct,
hedged, refused, off-topic, or truncated answer is NOT adequate.  If
a REFERENCE is provided, grade against it.  Be strict but do not
penalize harmless differences in format or extra correct detail.
```

The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first generated token's distribution:

```
[pre] I am going to show you a question.  Do NOT answer it.  Judge
honestly whether you yourself would answer it correctly.  Question:
{q} Would you answer this question correctly?  Reply with exactly one
word:  Yes or No.
[post] Here is a question and a proposed answer.  Question:  {q}
Proposed answer:  {a} Is the proposed answer correct?  Reply with
exactly one word:  Yes or No.
```

O Reproducibility

Representation-study sampling and splits use seed 42. Native serving uses sampled decoding; the main table contains one recorded run per query and arm. Later internal repeats are described separately in Appendix A.2. Core model environments and per-model pins are recorded in the experiment log (MiniCPM-o 4.5: PyTorch 2.8.0, Transformers 4.51.0; single-H100 instances). The revised benchmark figures, appendix rate table, and MiniCPM metric summary are generated from judged native rows by `paper/build_revision_figures.py`; its summary records input hashes, sample counts, scoring fields, and the timing and mixture definitions. The original main Table 1 is preserved and is not rewritten by this script; its NVDA block belongs to the separate replay experiment in Appendix D. Main thresholds are per-language calibration quantiles. Historical per-pool quantiles are confined to the experiments that explicitly use them. The audit snapshot `data/paper_revision_pr0907/` supplies the calibration manifest, OOF scores, thresholds, overlap exclusions, refit artifact, and benchmark file hashes. These support the score-level checks in Appendix A. The subsequent release through `3ade619` supplies the previously missing scripts and an input manifest. Its clean-checkout reproduction record reports rerunning export scripts 48/49 after hash-verifying 38 feature shards and 46 judged parquet files: five text exports are byte-identical and the OOF table is value-identical. Local revision checks independently verify the published OOF thresholds, validation ID hashes, arm gate hashes, and validation accuracy remixes; they do not rerun model inference or the full feature fit. The internal-stratum analysis is generated by `paper/build_internal_diagnostic.py` from the original query manifest and three native arms; its output records input hashes. It runs no models and estimates no new operating point.

P The native full-duplex regime

The earlier representation and harness experiments read the probe under a turn-based loop or its concurrent variant (Appendix I). The main benchmark instead uses the native regime. This section records its successive calibration and serving variants; they are distinct from repeated main-benchmark runs. MiniCPM-o 4.5 also ships a native full-duplex mode: audio prefills in 1 s units into the same context the model generates from, and the head itself emits ⟨listen⟩ or speaks, yielding the floor via its turn-end token. This is the deployed regime of the demonstration system, and this section recalibrates

Table 20: Native-regime probe AUC at the deployed listen→speak read point, under the OFFICIAL serving configuration (final deployed system; the chronological narrative above measured under inherited wrapper defaults). The merged refit adds the 2,700-row public-benchmark expansion (incl. 400 Mandarin rows) to the official core. En-4 mean .770 vs. turn-based .771: the English regime cost is eliminated; the Mandarin slice restores Reasoning-zh from the all-English chance collapse. The last column is the deployed gate: same 5,228 features, but every training label is the judged outcome of the same native-duplex answer the features were read from (instead of a separate deterministic turn-based answer). Label source is a wash for ranking (En-4 mean .771 either way; the paired bootstrap crosses zero on six of seven pools, TriviaQA $+.035$ [$+.006$, $+.065$] being the exception), so the deployed gate uses the in-regime definition. Internal test is scored against both the earlier harness labels and the native-judged labels of the same 240 queries.

pool	n	old cfg	official (2,542)	+exp3 (5,228)	+**native lbl. (deployed)**
internal (harness lbl.)	239	.830	**.846**	.844	.843
internal (native lbl.)	239	—	—	.878	.876
TriviaQA	250	.711	.759	.767	**.802**
WebQ	241	.736	.754	.772	**.777**
Llama-QA	245	.757	.789	**.815**	.807
SD-QA	200	.736	**.737**	.728	.700
Reasoning-zh	202	.606	.495	.605	**.621**

and re-validates the gate inside it. The read point needs no external end-of-turn detector: the chunk where the head first switches listen→speak is the commit-to-answer moment, and the probe reads there (the tail-k features then cover the model's first $\sim$8 answer tokens, a generation-side read the head hands us for free). For 87% of test queries this commit lands within $\pm$1 chunk ($\pm$1 s) of the true question end.

Recalibration. The full 2,310-row calibration mix was re-collected in this regime (fresh duplex session per query, features at the deployed read point, labels unchanged) and the same 12,288-d linear read refit. (The final deployed gate goes one step further and also re-labels: each training row's label is the judged outcome of the native answer produced in the same duplex session the features were read from, under the deployed sampling, rather than a separate deterministic turn-based answer. The two label sets agree on 80.8% of the 5,228 rows; the native failure rate is higher, .541 vs. .463, mostly turn-based-correct questions the sampled native answer gets wrong. Ranking is unchanged within noise, Table 20.) On native test-240 features the in-regime probe reaches AUC **0.830** [0.777, 0.878] (transferred concurrent probes: 0.807–0.818; turn-based reference 0.877). Externally the in-regime refit beats both transferred probes on all five pools: TriviaQA .711 (Δ+.080 [+.030,+.129]), WebQ .736, Llama-QA .757 (Δ+.118), SD-QA .736, Reasoning-zh .606 (Δ+.115); external mean .709. The regime ordering is monotone everywhere: turn-based (.877/.771 internal/external mean) $>$ native (.830/.709) $>$ concurrent (.689 external), i.e. a substantial share of the concurrent regime's cost was the harness read point, not liveness itself. Thresholds again do not transfer (balanced quantile 0.386→0.645); scores are decoding-temperature invariant (0.7-trained probe on temp-0.1 features: AUC 0.843).

Benchmark-overlap audit. The native benchmark contains known calibration overlaps. The precise collision counts, the reported deduplicated-refit sensitivity, and the distinction between that candidate and the gate actually used in the recorded arms are given in Appendix A. The offline audit does not establish overlap-free deployment or rule out pretraining contamination.

Validity. Re-mixing native local outcomes (gpt-judged, deployed decoding) with cached always-arm expert outcomes under the native scores (Table 21): under the deployed gate, gated accuracy clears matched-rate random on all five English pools at both deployed tiers: frozen .431→.669 at 49% escalation (p<.0001; ceiling .669), TriviaQA .568→.852 at 54%, WebQ .411→.680, Llama-QA .767→.841, SD-QA .455→.775 at 61%. In-regime training renormalizes the score distribution, so realized escalation rates sit near their nominal quantiles on the English pools (the earlier 64–73% drift was a transferred-threshold artifact). Reasoning-zh is an operating-point problem before it is a ranking problem: a single global threshold, quantiled on a 92%-English training mix, places the zh score distribution below every tier (1–2% fire at "balanced"), the language axis of Appendix I expressed as a fire-rate failure. The deployed fix needs no evaluation data: tier thresholds are quantiled on the training-set out-of-fold scores per language (en/zh), and the session selects the language's

Table 21: Native-regime gated accuracy under the official serving config and the deployed merged (official+exp3) gate (8ad remix: native scores select, native local outcomes + cached always-arm expert outcomes mix). p = permutation vs. matched-rate random; escalation rate in parentheses. Thresholds are the deployed per-language operating points: tier quantiles of the training-set out-of-fold scores, computed separately for the English and the Mandarin training rows (no evaluation pool is consulted); the Mandarin pool uses the zh thresholds. On frozen test the aggressive tier reaches the always-escalate ceiling at 49% of the expert cost. Reasoning-zh, which the global English thresholds never reached (1–2% escalation), now escalates at its nominal rates and beats matched-rate random at the two lower tiers. AlpacaEval (measured under the 2,542-row official gate) stays a clean honest negative.

pool	local	balanced	aggressive	always
frozen test	.431	.565 (27%, p=.0005)	**.669** (49%, p<.0001)	.669
TriviaQA	.568	.684 (20%, p=.0005)	.852 (54%, p<.0001)	.916
WebQ	.411	.527 (22%, p=.045)	.680 (58%, p=.020)	.809
Llama-QA	.767	.792 (4%, p=.002)	.841 (16%, p<.0001)	.910
SD-QA	.455	.605 (24%, p=.0035)	.775 (61%, p=.003)	.895
Reasoning-zh (zh thr.)	.525	.658 (34%, p=.023)	.713 (55%, p=.066)	.807
AlpacaEval (VB 1–5)	3.65	3.70 (6%, n.s.)	4.07 (37%, p=.24)	4.76

thresholds. The zh thresholds derived from the 400 Mandarin training rows alone land Reasoning-zh at 17/34/55% fire (nominal 15/30/50) and lift it from .525 to .609/.658/.713, beating matched-rate random at the two lower tiers (p=.0065, p=.023). This static per-language point is only as good as the match between the language slice's family mix and the deployment stream: when the Mandarin training slice is widened from 400 to 1,900 rows with higher-failure families (below), its balanced quantile moves from .39 to .75 and fires 5% on Reasoning-zh; a family-balanced quantile recovers 13/24/38%. The composition-free alternative (each pool's own score quantile, or its online form, a 100-turn sliding-window tracker with no labels and no pool identity) realizes nominal rates on every pool and converges within $\pm$.06 of nominal on the English pools.

Is the Mandarin gap a language gap? No: it is a task-type gap. A targeted 1,500-row Mandarin expansion (Chinese SimpleQA long-tail facts, C-Eval and CMMLU knowledge MC, the remaining XCOPA-zh; public benchmarks, deduplicated against every pool, judged on the native answers; native failure .578) does not move Reasoning-zh when added to the calibration mix (.621$\rightarrow$.627, paired $+$.006 [$-$.026, $+$.040]; internal test $-$.010, English external $+$.006), and Mandarin-only training is worse everywhere (Reasoning-zh .591, internal .758). Read the other way, the deployed gate scores these 1,500 unseen Mandarin questions at AUC **.811** [.789, .833] (long-tail .719, C-Eval .730, CMMLU .642, causal .693), the same band as the English pools. What the probe does not read is multi-step reasoning, which is what Reasoning-QA is; the ".62 on Mandarin" number is that gap wearing a language label. The expansion is therefore not deployed and serves as a second, larger Mandarin held-out pool. (A side observation from the same dump: on the causal-commonsense family 34% of the native answers switch to English while remaining judged adequate, a language-consistency defect the adequacy label does not see.)

Reasoning coverage and in-context rows: the read point saturates. Two further targeted sets test the two remaining hypotheses. A bilingual multi-step reasoning pool (1,319 rows: LogiQA and LogiQA-zh, MATH levels 4–5 without LaTeX, StrategyQA, BBH date/deduction/causal/temporal/navigation, CMATH grades 4–6, Ape210K; native failure .595) leaves Reasoning-zh unchanged when added to calibration (.621$\rightarrow$.620, paired $-$.002 [$-$.039, $+$.038]) and moves nothing else beyond noise. The reason is visible inside the pool itself: cross-validated AUC on the reasoning families is only .61–.76 (overall .736 [.708, .762]), against .77–.82 for fact, knowledge-MC, and causal families. Whether a multi-step chain will come out right is weakly encoded in the listen$\rightarrow$speak commit state, which precedes the chain; the .62 on Reasoning-zh is that ceiling, not a coverage gap. Second, the calibration questions re-collected as the second turn of a session (a spoken carrier question and its answer first) fail more often (.517$\rightarrow$.620; 26% of labels flip) while the deployed gate scores them lower (mean shift $-$.097): at the balanced tier the in-context escalation rate halves (.252$\rightarrow$.115) although ranking holds (AUC .861 [.815, .896]). Adding these rows to training does not change AUC either; the defect is the operating point, and the deployed demo therefore thresholds on a sliding-window quantile of the session's own recent onset scores (100-score window, static per-language threshold until 20 scores exist), which follows such shifts without labels or pool identity. After a label-definition fix and three targeted expansions (≈4,000 rows), external ranking sits where the 5,228-row gate left it. The head is not the bottleneck either: on the same features, per-dimension standardization costs .07 of English external AUC (the raw activation scale is informative), an elastic-net head is a wash (internal +.008, external −.007), and non-linear heads are clearly worse (256-unit MLP .702, PCA-256 + gradient-boosted trees .715, vs. .771). Nor are several heads: a three-way task-type router trained on the same features is 97% accurate, yet no family-specialized head beats the single head on its own family (facts .846 vs. .847, reasoning .761 vs. .766, chat .672 vs. .720), and the routed mixture is flat-to-negative externally (SD-QA −.033 [−.060, −.007]). What the commit-chunk state encodes about the coming answer is extracted by one linear read; the remaining headroom lies in what is read: later into the answer, other layers, or a second signal family such as verbalized self-evaluation (Table 9), not in how. The talker's own local floor is lower in-regime (frozen .483→.431 under the official serving configuration; the inherited-defaults harness measured .371, and the ≈one-third-temperature / ≈8-point decomposition belongs to that older figure), which raises the gate's marginal value in deployment. The in-regime calibration scaling curve (measured under the default serving configuration) mirrors the concurrent one and is not saturated: internal .717→.830, external mean .643→.709 from 360→2,310, and a further public-benchmark expansion to 5,009 rows (2,300 English across the same seven families + 400 Mandarin: MGSM-zh and XCOPA-zh through the same TTS pipeline, source-disjoint from Reasoning-zh) continues the $\approx +.02$-per-doubling external trend to **.736** while internal stays flat (.834), and the Mandarin slice moves the axis English data never touched, Reasoning-zh .606→.659 (Δ vs. conc-2310 +.169 [+.085,+.251]; the deployed official-configuration merge reads the same pool at .605 with turn-based labels and .621 with the deployed native labels, Table 20). Two caveats on reading this curve: its six points are the project's chronological accumulation, each step adding new query families, coverage, or a serving-configuration change, not repeated random subsamples of the final set, so it evidences that added coverage transfers, not that another random doubling of the same families would return +.02; and the internal/English-external plateau against the turn-based reference (.771) is an empirical baseline at n≈200–250 per pool, not a ceiling.

Ceiling, stability, and the pre-answer read. Three diagnostics on the deployed 5,228-row gate. (i) **Ceiling.** A pool-prior scorer (each query scored by its training pool's out-of-fold failure rate) reaches AUC .786 against the probe's .848 on the training out-of-fold scores, and .767 against .876 on frozen test; AUC restricted to same-pool pairs is .743 (train) and .820 (test). Per-question discrimination is thus +.06–.11 of AUC on top of the type prior, the same decomposition as the n=600 audit of Section 3.1; at a 30% budget the prior alone recovers .475 failure recall against the probe's .506 (random .299). The external pools, each a single source, measure the within-pool component directly (.70–.81, Table 20). (ii) **Stability.** An independent second native pass over frozen test (deployed sampling, T=0.7, its own judged label) changes 98% of answer texts and flips 14% of native labels, yet the read at the commit chunk barely moves: feature cosine .95, score correlation .95, and the tier decisions agree on 91–94% of queries. Each run's features predict the other run's outcome about as well as their own (.848/.887 vs. .876/.872), so the probe reads a property of the question in context rather than the fate of one sampled answer; against the 204 queries whose two outcomes agree, AUC is .91–.93, which bounds the single-sample label noise in the reported numbers. (iii) **Pre-answer read.** The same layer-22 read taken before the onset chunk's generate call (no answer tokens in the tail), scored by the deployed head without refitting, gives AUC .861/.837 against the two label sets, .011–.015 below the deployed read; the first ≈8 answer tokens are worth about one AUC point, and the pre-answer variant needs its own operating points (the deployed quantiles fire 0/1/20% on it). A hard-example check closes this route from the training side: at fixed regularization, upweighting the 711 calibration failures the out-of-fold probe misses at the widest tier drops external transfer from mean AUC .743 to .715 (4×), below every draw in 100-subset random-failure null distributions, count-, weight-, and source-composition-matched (null means .737/.740, empirical $p < .01$) — the harm is specific to those rows, not to extra positive mass. Retuning the regularizer per arm recovers most of the loss (.735 at 4×, with C shrinking $3{\times}10^{-4} \rightarrow 3{\times}10^{-5}$ at 8×): the solver's best response is to suppress the fit to the upweighted rows, and no weighting improves on the unweighted baseline. The missed failures do not benefit from emphasis under the tested weighting and regularization settings at this read point. This is consistent with limited linearly readable signal, but does not by itself distinguish representation overlap from label noise or finite-sample effects. Related training-dynamics studies show that ambiguous and mislabeled examples can affect generalization

[Swayamdipta et al., 2020, Pleiss et al., 2020]; their results do not establish the mechanism in our frozen speech representations. Cheang et al. [2025] find that associated hallucinations overlap with factual outputs in hidden-state space, reducing the effectiveness of the tested detectors on factual completion tasks. David [2025] finds that eventual correctness is predictable from as few as four reasoning tokens in math tasks, without testing a zero-token baseline. These findings motivate our read-point analysis, rather than establishing that correctness cannot be predicted before generation.

Live validation. Full live runs on frozen test-240 (real raw-audio ASR uplink, real expert, wait paced at one chunk per second) validate the remix arithmetic where the channel is thin and expose where it is not: balanced delivers .504 at 23% fire (remix predicted .492 at 24%); aggressive delivers .537 at 45% against a .596 remix prediction, and the gap decomposes entirely onto fired turns: live fired-turn accuracy .495–.589 vs. the .667 cached-expert ceiling. The native relay channel (the answer re-entering as a text unit and being voiced) is the regime's lossy element (89% of relay units need the nudge retry), playing the role the transcription uplink played in the turn-based loop; the uplink itself is now clean (raw-audio ASR, p50 0.9 s). Latency, expert-cache warm: expert p50 4.4–5.7 s, wait p50 5–6 chunks, fire$\rightarrow$relay first audio p50 $\sim$7–8 s. Selection is visible live: unfired subsets score .478/.573 against the .371 unconditional local floor.

Dialogue-act gating. Full duplex creates commits that are not answers: after a barge-in or a backchannel the head speaks to manage the floor ("Okay."), and the failure probe is out-of-distribution there, on 194 TTS'd floor-management stims (stop commands, backchannels, acknowledgments, fillers; en+zh) the unmodified gate false-fires on 37% at the aggressive tier (stop commands 45%), i.e. it would escalate "stop" to the expert. The fix is a second linear head on the same L22 read, info-seeking vs. floor-management, which separates the two acts perfectly (OOF AUC 1.000); thresholded at the question-side 0.5th percentile it costs 0.5% of true escalations and admits 0% of floor stims. Escalation requires both heads: $\mathrm{act} \geq \tau_a \wedge P(\mathrm{fail}) \geq \tau$. The deployed demo answers floor turns naturally and reports the act score per commit; the failure probe's calibration is untouched.

Live use then sharpened both heads within minutes: scripted pools miss what ten minutes of real conversation finds. (a) Live mic speech shifts both act distributions into the calibration gap (real request-phrased questions score $\sim$.93, live floor turns $\sim$.37 vs. a standalone-stim maximum of .05): a percentile-anchored τ_a rejected a real question and the talker delivered an empty promise. The fix is a gap-center threshold plus in-context calibration stims, each utterance dumped as the second turn of a carrier conversation, matching how floor acts actually arrive; the act read stays perfectly separable (AUC 1.000) under both conditions, and in-context stop commands false-fire the failure probe on 57% at the aggressive tier, so the act term is not optional. (b) The native refit had silently dropped the FreshQA real-time extension ("the stock price of NVDA today" scored $P(\mathrm{fail}){=}.29$ and stayed local); replaying that recipe in-regime restores fast-changing heldout fire to .45/.75/1.00 across tiers at zero internal-AUC cost (.830$\rightarrow$.833), with budgets still quantiled on the core mix. (c) A relay's own speak-onset must not re-enter the gate (re-fire with an empty uplink); deliveries are guarded to their end of turn. The pattern across all three: the mid-layer signals separate cleanly every time, and what re-opens at each regime or domain shift is calibration coverage and thresholds: interactive use is itself an evaluation modality.

The same lesson recurred with *phatic* questions. Utterances that are question-shaped but carry no information need ("How are you?", "Can you hear me?") pass an act gate trained on floor-*command* negatives—grammatically they are questions—and the failure probe is again out-of-distribution: on 88 TTS'd phatic stims (social openers, channel checks, assistant-meta; en+zh, two voices) it false-fires on 97–100% of social and channel-check stims at the aggressive tier (37–44% at balanced), i.e. it would escalate "how are you" to the expert, and live users hit exactly this. Adding the phatic stims to the act-gate negatives closes the category at no cost: the act read stays separable (OOF AUC 0.9999), all phatic categories score below threshold (0% admitted, residual false-fire 0%), and the question-side cost *drops* from 0.5% to 0.24% of true escalations—the wider negative manifold moves the joint-gap threshold to a better operating point.

Multi-turn grounding. A stateless expert uplink breaks on follow-ups: "what is NVDA trading at" then "what about Apple" sends only the second utterance, unresolvable in isolation. The probe itself is unaffected: it reads L22 with the prior turns live in the model's KV cache, so the fix is in the uplink, not the gate: the deployed demo threads a rolling resolved-dialogue history into the expert input and asks it to resolve references, after which the follow-up correctly returns Apple's share price. The measured eval pools are single-turn, so this is a deployment capability rather than a table; systematic multi-turn evaluation is future work.

Floor control, natively. With the harness gone, 8ba's question re-poses on the head itself (108 scripted overlap trials, stims spoken over the talker at fixed offsets). During answer speech, backchannels almost never stop the floor (false-stop 3/24 within a 6-chunk window) with no ASR or lexicon anywhere in the loop; sustained question speech yields the floor (6/12 within 6 chunks, median 8.5 post-stim); short burst commands ("Stop!") mostly do not (3/12): the native head reads sustained speech, not commands, a naturalness-for-reliability trade the old energy-VAD did not make. The escalation machinery is floor-transparent where it should be: relays complete under overlapping stims (26/30), and backchannels during the thinker wait leave the pending relay intact (8/10). The weak stop-command sensitivity decomposed into two causes, found in sequence. **Mechanism**: instrumenting the serving loop shows the head never samples $\langle$listen$\rangle$ mid-turn (zero attempts across 36 trials), so its only stop channel is the turn-end token. **Configuration**: a user-demanded vanilla control (the bare official serving stack, no gate machinery) revealed that our loop had inherited the checkpoint wrapper's sampling defaults (k=100) rather than the official demo's serving config (k=20, a 3-chunk startup listen guard, an assistant-style system prompt). Under the official config the turn-end channel is responsive: a spoken "stop" two seconds into an enumeration answer ends the turn in 2.0–2.2 s in 3/3 scripted sessions, versus 1–13 s (once running the answer to completion) under the inherited defaults: wide-k sampling dilutes away the rising turn-end probability. The deployed demo now serves the official config, and the earlier floor-control stop numbers measured under the inherited defaults should be read as a lower bound on native responsiveness. Two durable lessons: serving-parameter parity with the reference stack is part of "native", and a vanilla control arm belongs in the harness from day one. The one true interaction: a new question during the wait derails the pending relay 7/10 times: the head simply moves on, which is the empirical case for aborting the in-flight expert call on user speech, left as deployment future work. Latency: listen chunks cost 0.04 s and speak chunks $\sim$0.5 s on one H100 (realtime holds); fire$\rightarrow$stall-audio 0.2–1 s; fire$\rightarrow$relay-first-audio is the expert wall clock (13–20 s observed) plus $\sim$2.5 s.

P.1 Failure type and later-read diagnostics

A separate audit of the six QA pools classified never-arm failures using question text, an audio transcript, the reference answer, the local answer, and the judge note. The reported specific-but-wrong class contains 418 examples (about 70% of logged failures) and is ranked at AUC 0.813 against correct answers. Perception errors ($n = 36$) reach 0.889; execution slips ($n = 44$, including 35 Mandarin reasoning cases) reach 0.469. The class labels are model judgments, not direct observations of an internal failure mechanism. A fluent answer to a misheard question can be assigned to the wrong class.

A fresh read-time dump captured five L22 vectors during each native generation and fitted a separate probe at each point (5,236 judged calibration rows, row-random five-fold fitting, $C = 3 \times 10^{-4}$). These are different sessions and calibration rows from the main arms. The pre-onset vector precedes the onset generation call; the onset vector follows it. Later vectors follow one, two, or three additional answer chunks. The last available vector pads sequences that finish earlier (0.2/2.8/10.6% at the three later reads). Table 22 preserves the corresponding diagnostic.

Table 22: Separate native read-time sweep: failure-prediction AUC. External macro equally weights five speech-QA pools. This evaluates representation availability during generation, not a live deferred routing policy or its audible latency.

Read point	Calibration OOF	Internal test	Reasoning-zh	External macro
Before onset	.848	.832	.711	.753
At onset	.850	.852	.648	.755
After 1 chunk	.852	.854	.681	.762
After 2 chunks	.855	.846	.769	.791
After 3 chunks	.857	.858	.744	.801

The sweep supports a timing tradeoff: later states can improve failure prediction, particularly on reasoning, but may require withholding or revising generated content. A cached two-stage mixture improves external mean answer-content accuracy from .711 to .718 at a 30% budget. This is a diagnostic remix rather than a newly recorded two-stage serving arm.